Schedule

Monday 12 January

9:15 – 9:30	Introduction
9:30 – 10:15	Variant to gene to function
10:15 – 10:30	Break
10:30 – 11:15	AI for genomics
11:15 – 12:30	Work on mini-project
12:30 – 13:30	Lunch
13:30 – 17:30	Work on mini-project

Wednesday 14 January

9:15 –10:15	Single-cell genomics
10:15 – 10:30	Break
10:30 –11:30	Functional genomics using single-cell RNA-seq
11:30 – 12:30	Work on mini-project
12:30 – 13:30	lunch
13:30 – 15:30	Work on mini-project **(hand-in before 15:30)**

Single-cell and spatial genomics

Ahmed Mahfouz

Dept. of Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

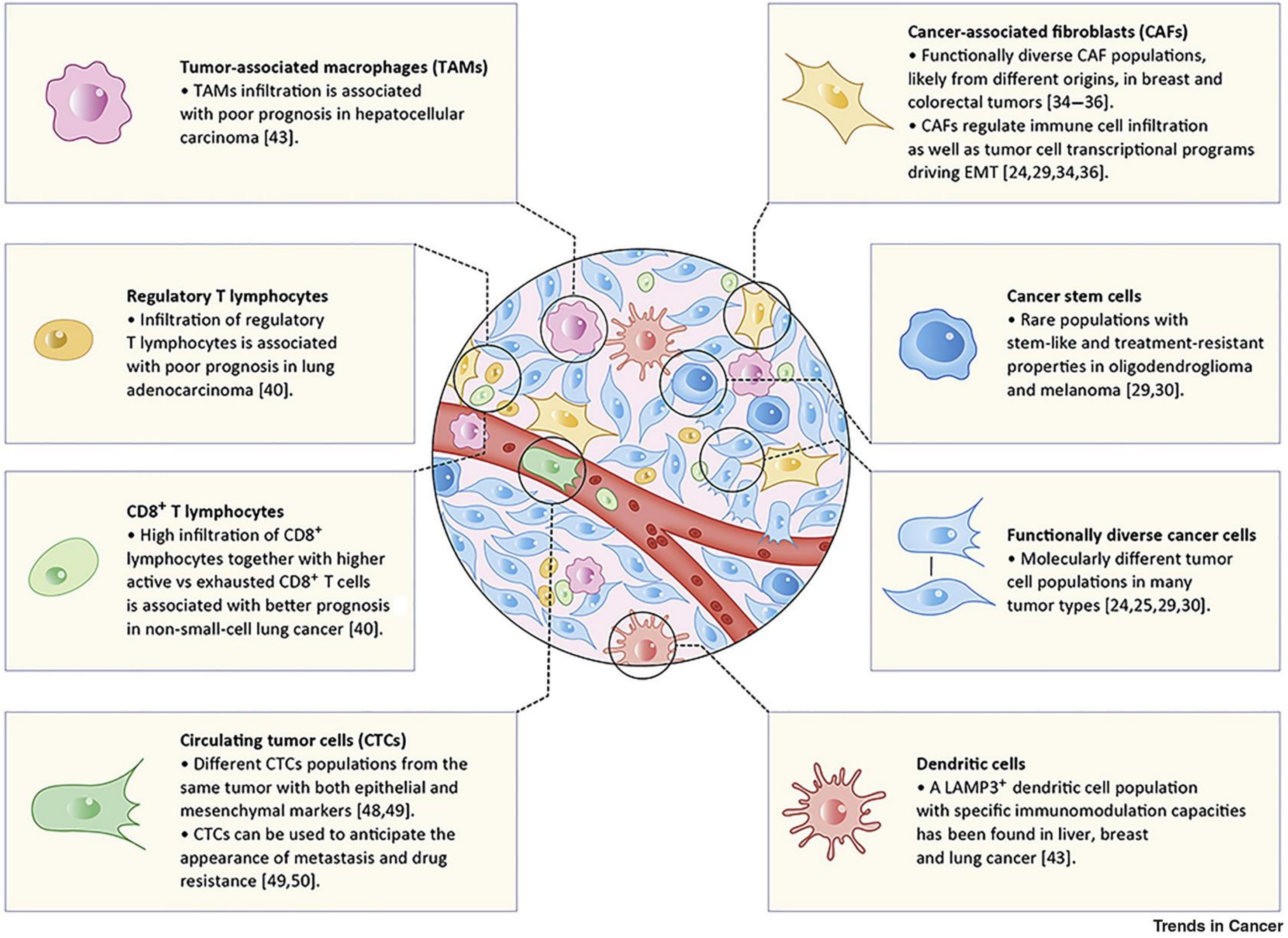
RNA sequencing

Input needed ≈ 10 – 100 ng mRNA

Each cell has ≈ 0.4 – 1 pg mRNA

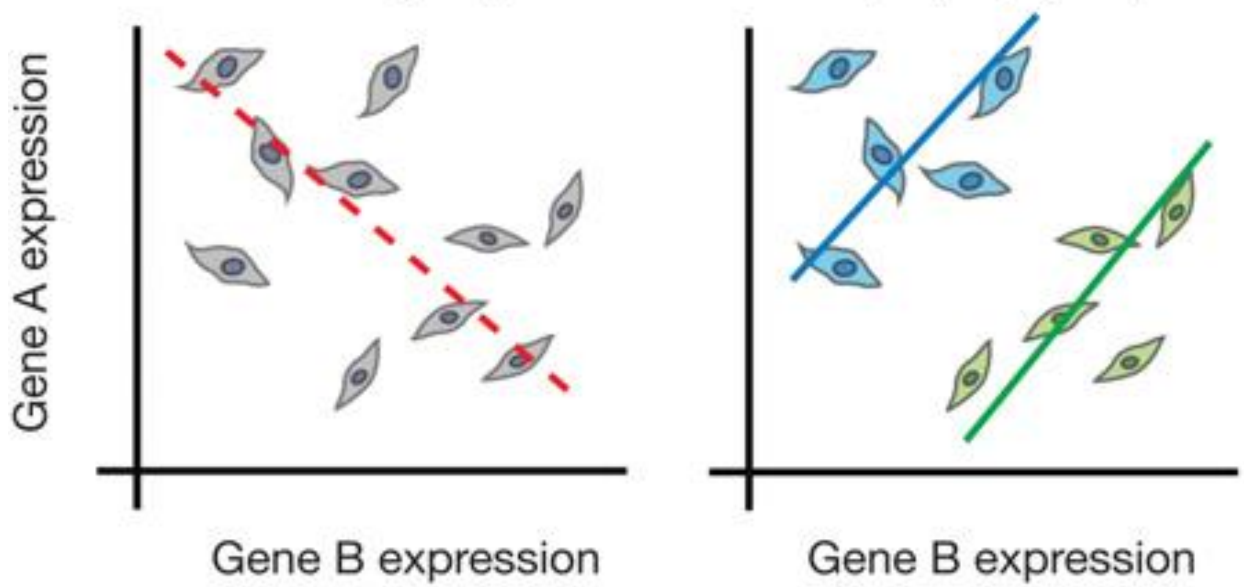
Solution: **pool mRNA from multiple cells** + PCR

In many cases, “bulk” profiling is insufficient



Silva et al., *Trends in Cancer* 2020

Simpson's Paradox



Simpson's Paradox describes the misleading effects that arise when averaging signals from multiple individuals.

Trapnell (Genome Research 2015)

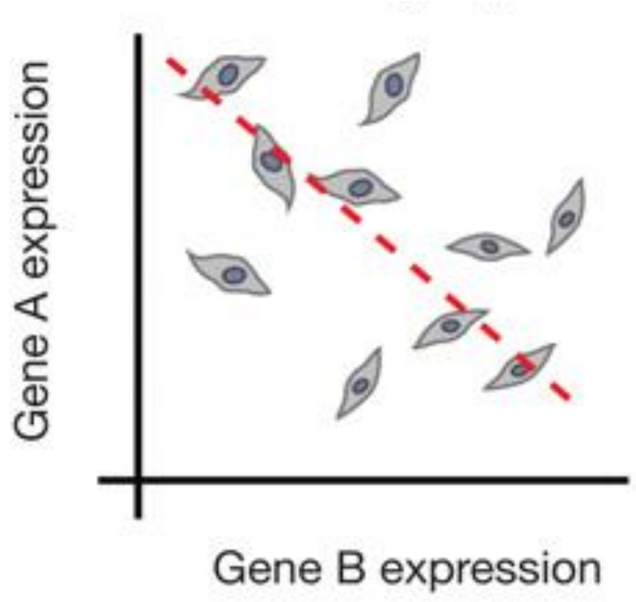
Bulk profiling obscures tissue heterogeneity

Bulk genomics

Single-cell genomics

Credits: Bo Xia (NYU)

Why single cells?



Simpson's Paradox describes the misleading effects that arise when averaging signals from multiple individuals.

Trapnell (Genome Research 2015)

Traditional technologies for single-cell analysis

In living cells

Method	Species?	Endogenous?	Real-Time?	# probes?	Amplification	Advantages	Disadvantages
MS2 or Spinach	RNA	No	Yes	<5	No	Spatial Info	Need long UTRs
Molecular Beacons or SmartFlares	RNA	No	Yes	<5	No	Spatial Info Lots of cells	Requires microinjection

In dead cells

Method	Species?	Endogenous?	Real-Time?	# probes?	Amplification	Advantages	Disadvantages
FISH	DNA, RNA	Yes	No	<5	No	Spatial Info Lots of cells	Expensive
In-Situ Sequencing	DNA, RNA	Yes	No	Many	Yes	Spatial Info Lots of cells	Bias Slow
Single-cell (RT)-PCR	DNA, RNA	Yes	No	<500	Yes	Simple	Need to know targets Bias
Single-cell Sequencing	DNA, RNA	Yes	No	Many	Yes	Genome wide	Bias Expensive

Slide: Mannolis Kellis (MIT), adapted from Raj & van Oudenaarden (2009)

Traditional technologies for single-cell analysis

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Molecular Beacons or SmartFlares	RNA	No	Yes	<5	No	Spatial Info Lots of cells	Requires microinjection

+ Dynamics
+ Spatial
– Few/known genes

In dead cells

Method	Species?	Endogenous?	Real-Time?	# probes?	Amplification	Advantages	Disadvantages
FISH	DNA, RNA	Yes	No	<5	No	Spatial Info Lots of cells	Expensive
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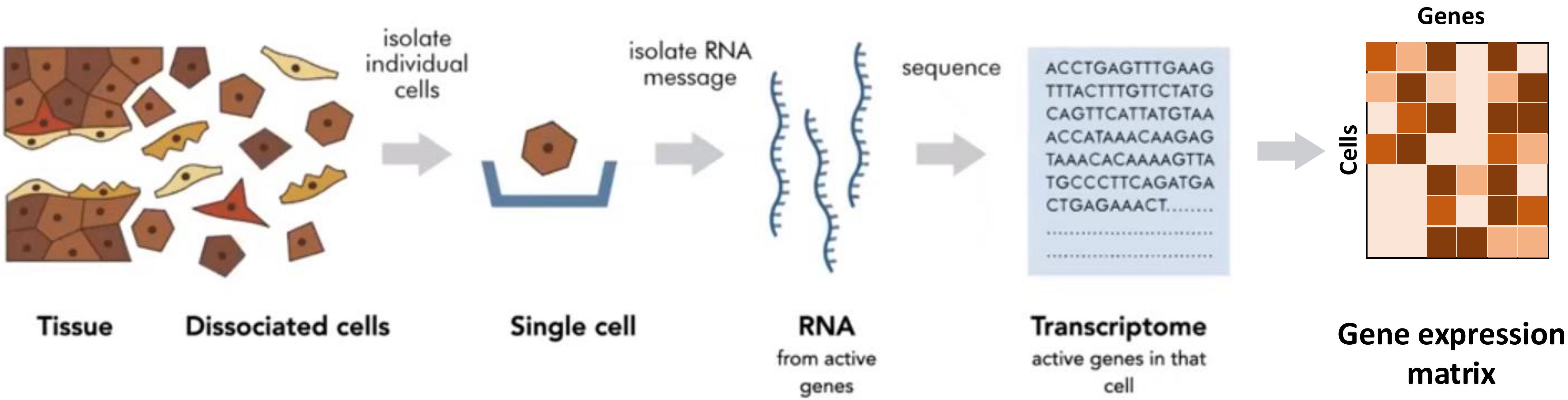
Single-cell (RT)-PCR	DNA, RNA	Yes	No	<500	Yes	Simple	Need to know targets Bias
Single-cell Sequencing	DNA, RNA	Yes	No	Many	Yes	Genome wide	Bias Expensive

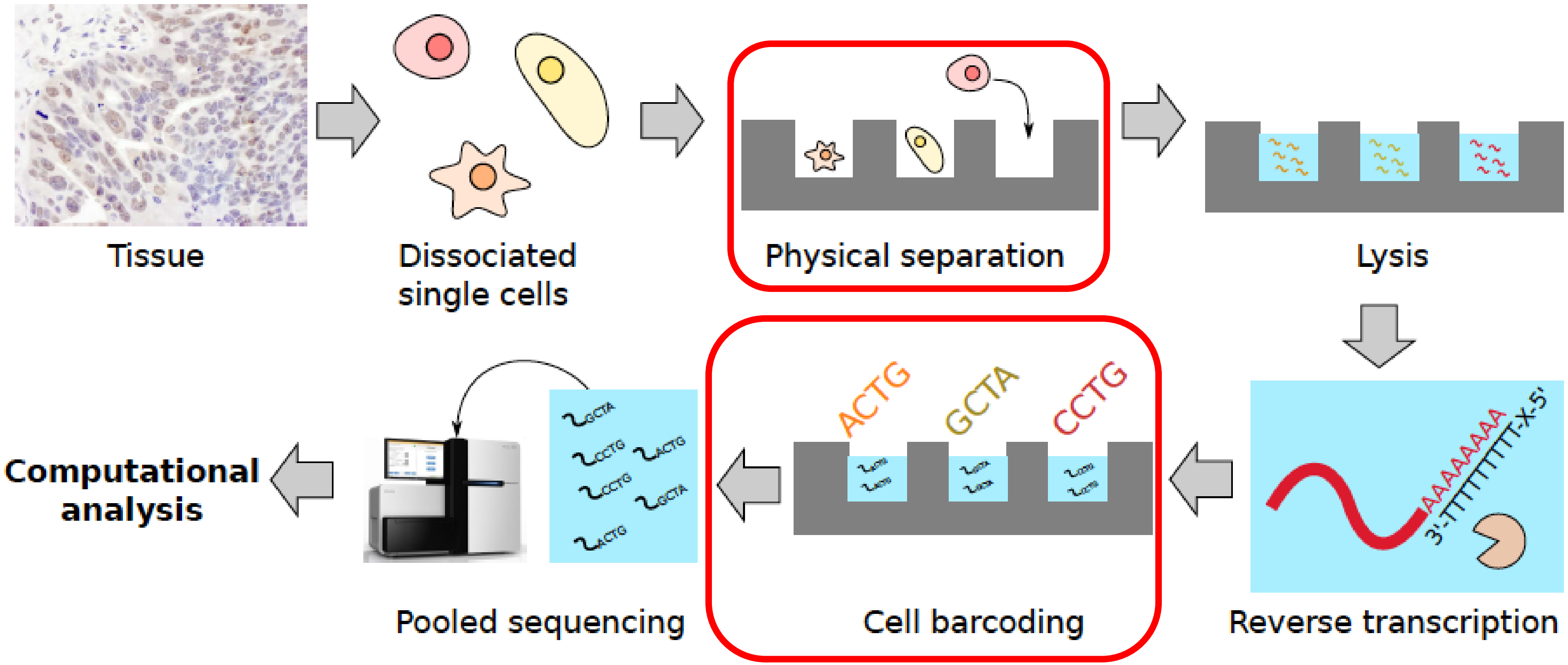
+ More genes
– No dynamics

Slide: Mannolis Kellis (MIT), adapted from Raj & van Oudenaarden (2009)

Single-cell RNA sequencing (scRNA-seq)

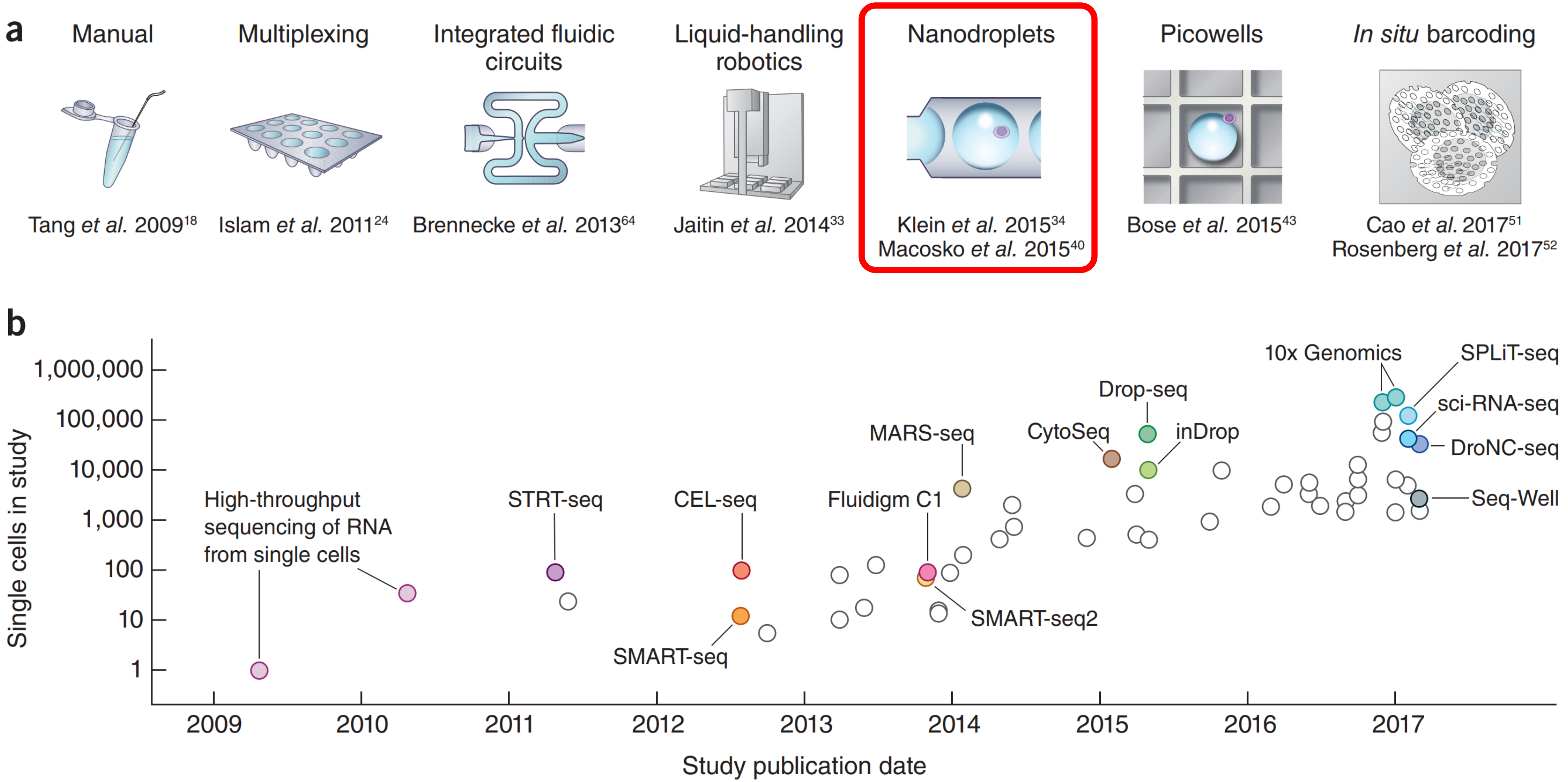


Single cell RNA-sequencing (scRNA-seq)



Aaron Lun (Marioni Group, EMBL-EBI)

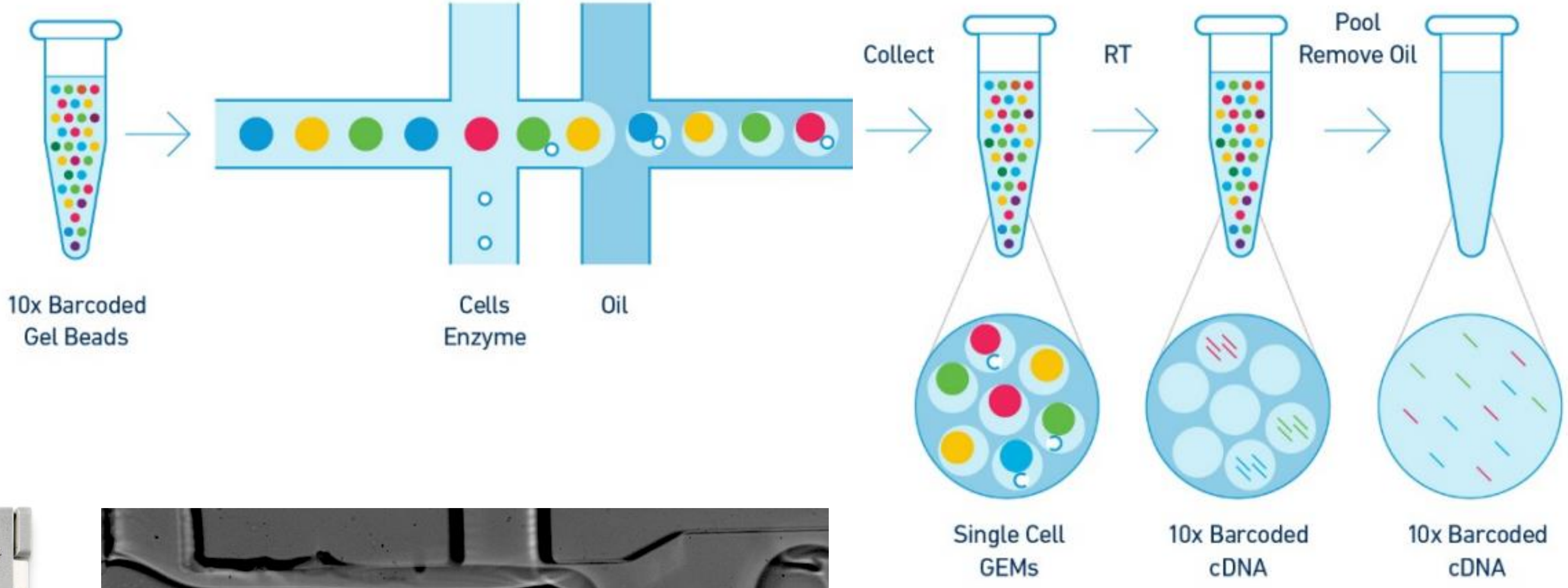
Exponential increase in throughput



Svensson et al. Nature Methods 2017

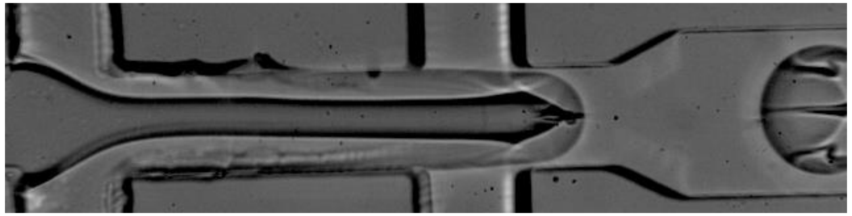
Droplet-based scRNA-seq

10x Genomics
(Chromium Controller)



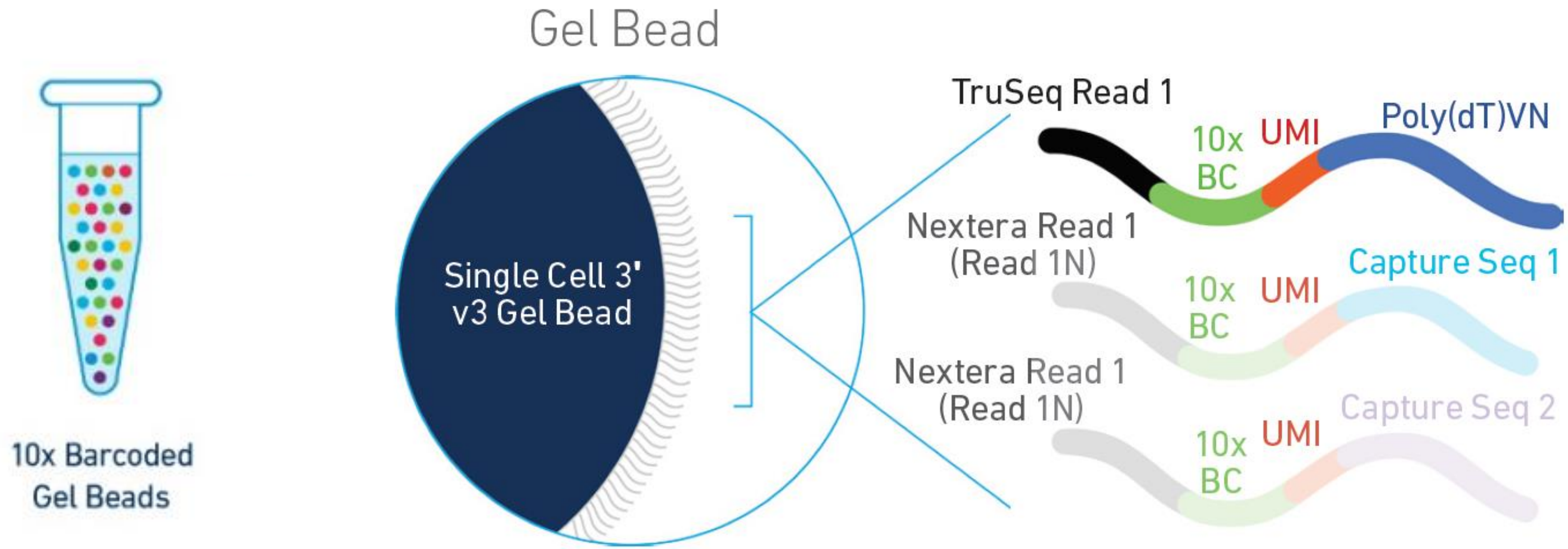
Single-use microfluidics chip

Recovery well
Gel Bead well
Sample well
Oil well



10x Genomics

Droplet-based scRNA-seq



10x BC (16bp) Unique for each bead
UMI (10bp): unique for each mRNA molecule

10x Genomics

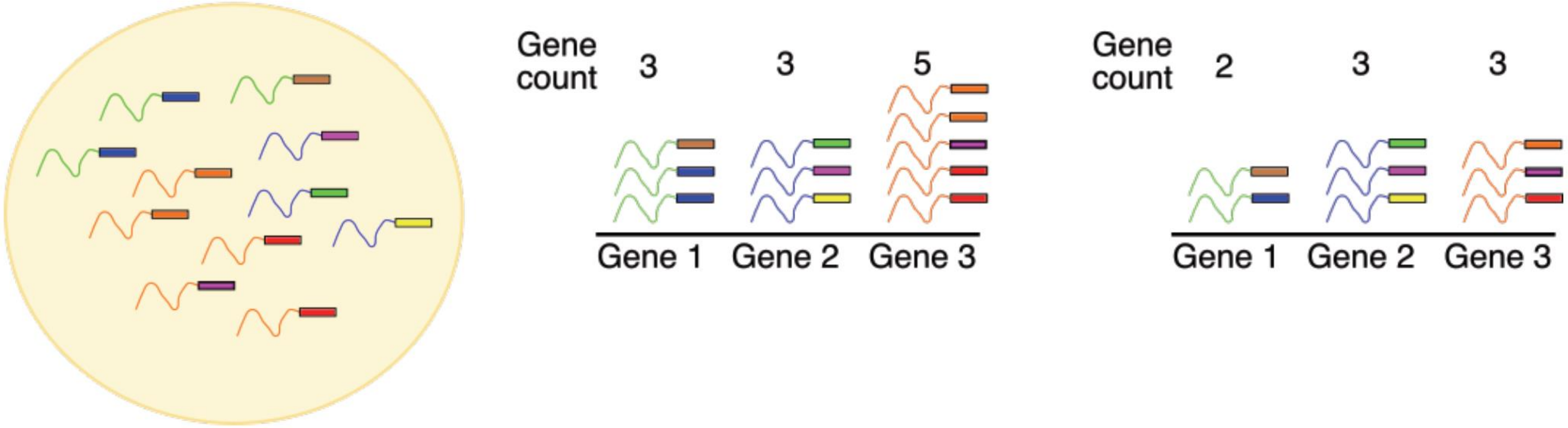
Unique Molecular Identifiers (UMIs)

- UMIs reduce the amplification noise by allowing de-duplication of sequenced fragments

Sequenced fragments from an individual cell

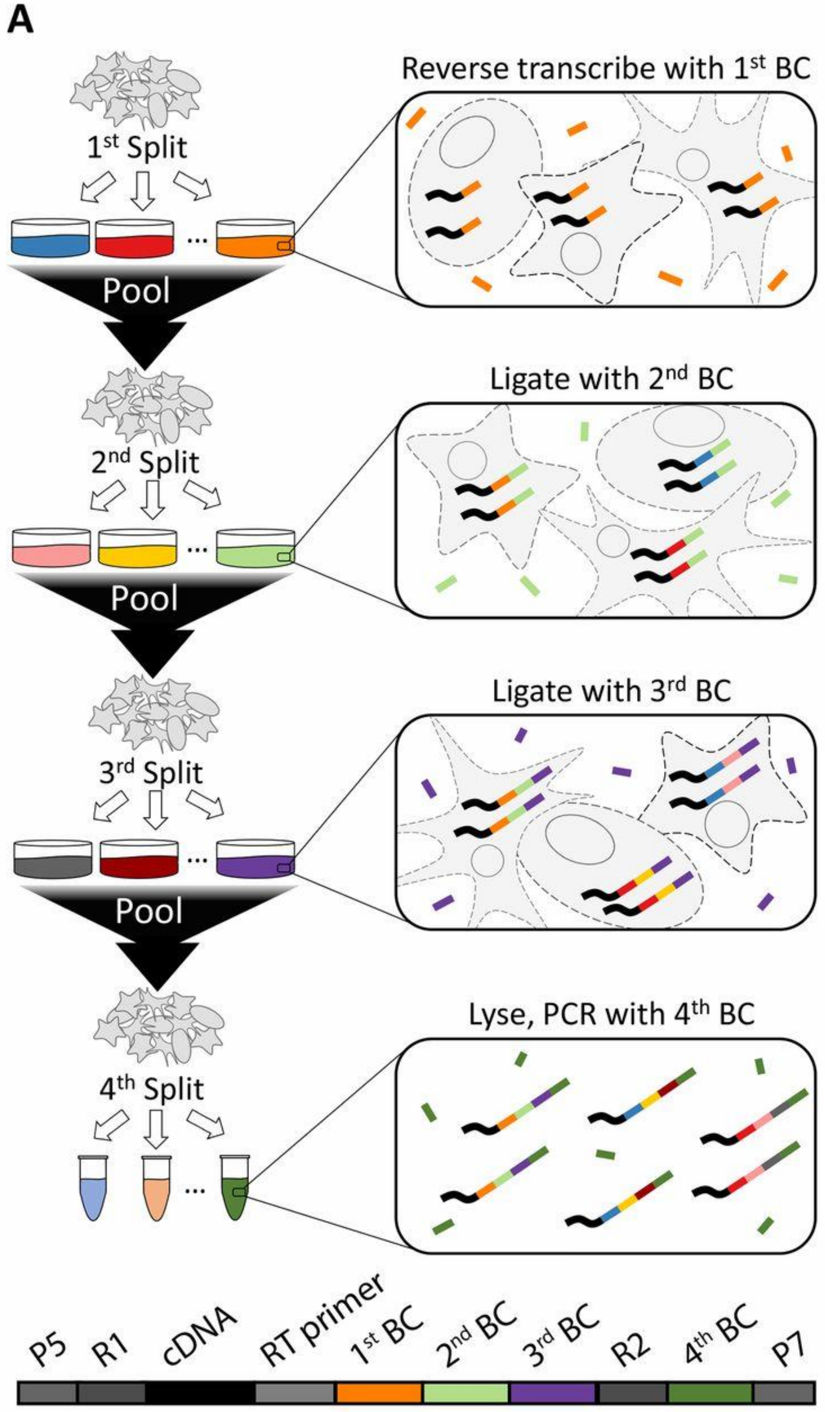
Pre de-duplication

Post de-duplication



Split-seq

- Single cells never individually isolated
- Instead: fixed, and mRNA is manipulated in situ inside each cell
- Split cells into ~100 wells (e.g. 96 or 384-well plate) with unique barcodes in each well
- Labels all cells with a first barcode, for that well. Chance of same barcode: 1/100
- Pool cells, shuffle, split again, randomly re-assorting into same set of ~100 wells
- Add second barcode. Chance of same 2 barcodes: 1/10,000.
- Repeat: pool, shuffle, split, add 3rdbarcode. Chance of same 3 barcodes: 1/1,000,000
- Can scale number of cells exponentially by number of barcoding rounds

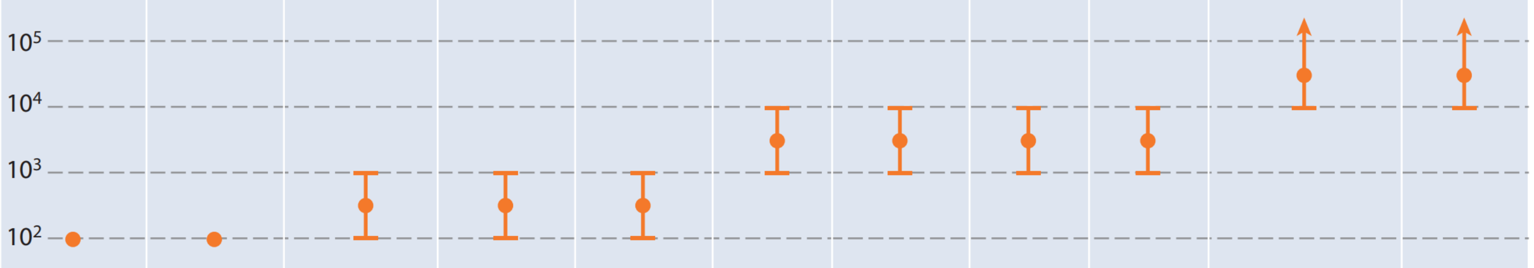


Rosenberg et al. *Science* 2018

Popular scRNA-seq protocols

	SMART-seq2	CEL-seq2	STRT-seq	Quartz-seq2	MARS-seq	Drop-seq	inDrop	Chromium	Seq-Well	sci-RNA-seq	SPLiT-seq
Single-cell isolation	FACS, microfluidics	FACS, microfluidics	FACS, microfluidics, nanowells	FACS	FACS	Droplet	Droplet	Droplet	Nanowells	Not needed	Not needed
Second strand synthesis	TSO	RNase H and DNA pol I	TSO	PolyA tailing and primer ligation	RNase H and DNA pol I	TSO	RNase H and DNA pol I	TSO	TSO	RNase H and DNA pol I	TSO
Full-length cDNA synthesis?	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes	No	Yes
Barcode addition	Library PCR with barcoded primers	Barcoded RT primers	Barcoded TSOs	Barcoded RT primers	Barcoded RT primers	Barcoded RT primers	Barcoded RT primers	Barcoded RT primers	Barcoded RT primers	Barcoded RT primers and library PCR with barcoded primers	Ligation of barcoded RT primers
Pooling before library?	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Library amplification	PCR	In vitro transcription	PCR	PCR	In vitro transcription	PCR	In vitro transcription	PCR	PCR	PCR	PCR
Gene coverage	Full-length	3′	5′	3′	3′	3′	3′	3′	3′	3′	3′

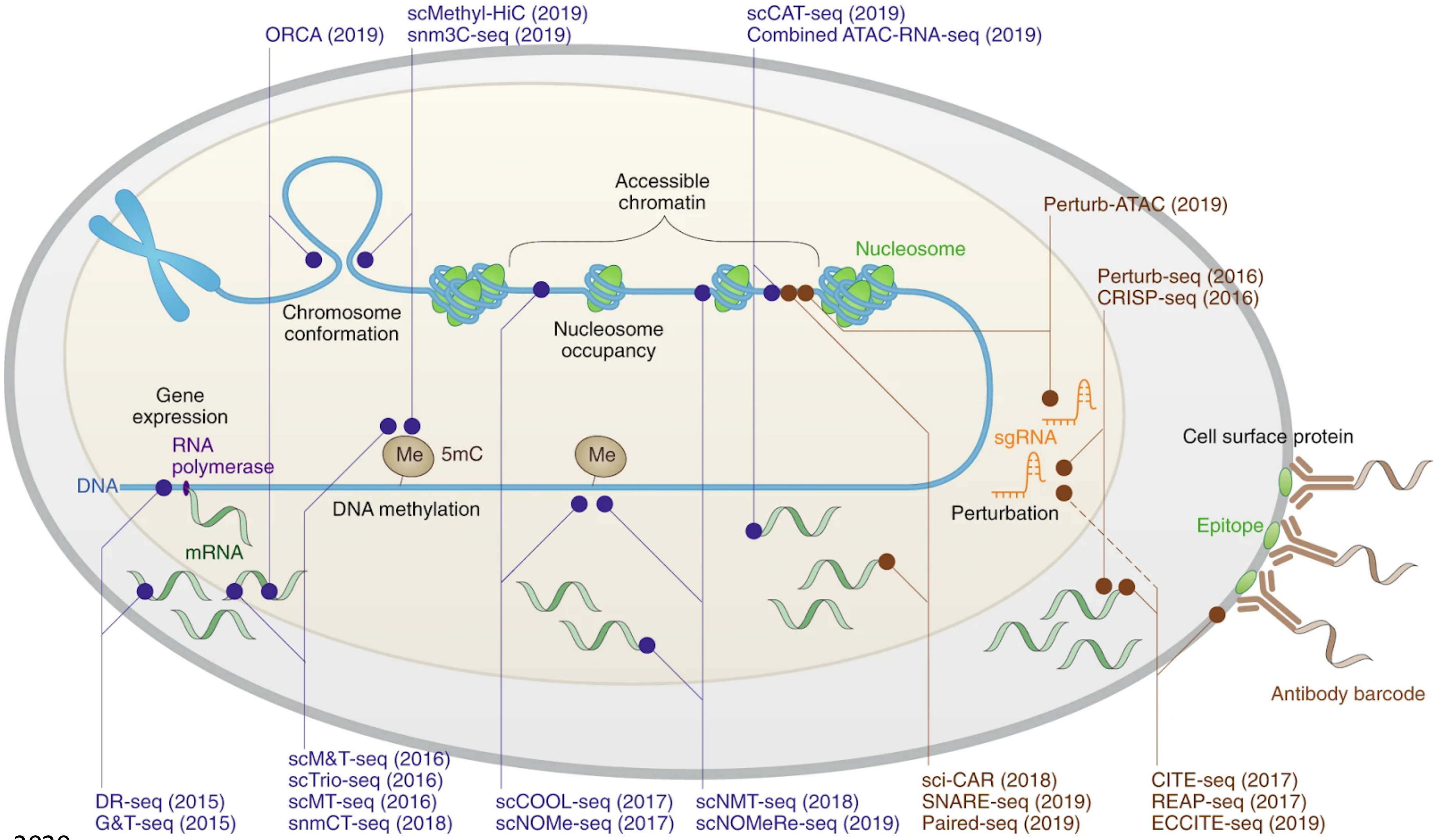
Number of cells per assay



Key points:
- Full transcriptome vs 3’ vs 5’
- Throughput varies
- Automation varies
- Cost varies

Chen et al. 2018. DOI:10.1146/annurev-biodatasci-080917-013452

Single-cell multi-omics (not only mRNA!)



Zhu et al. Nat Methods 2020

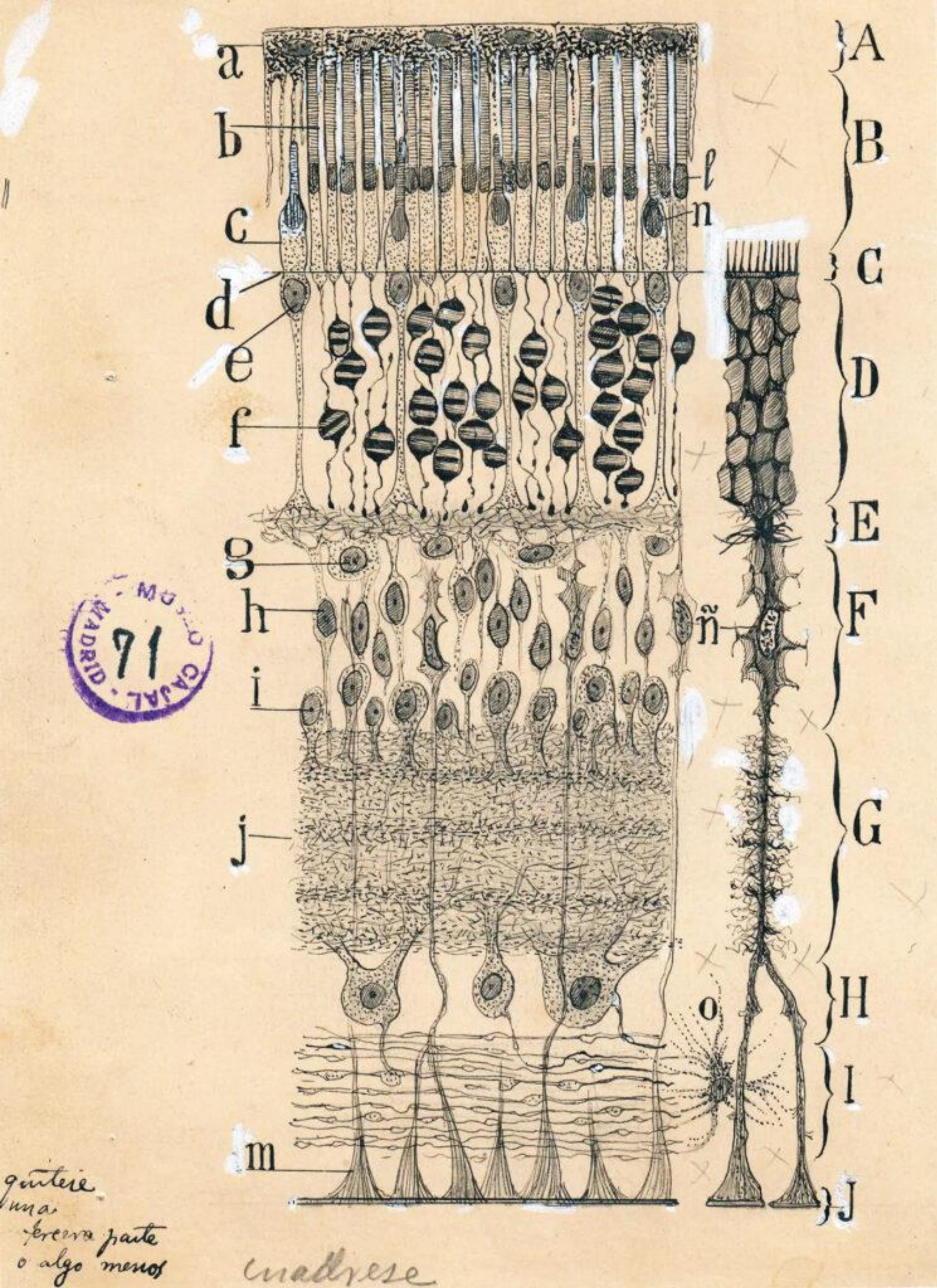
Single-cell profiling obscures tissue organization

Bulk genomics

Single-cell genomics

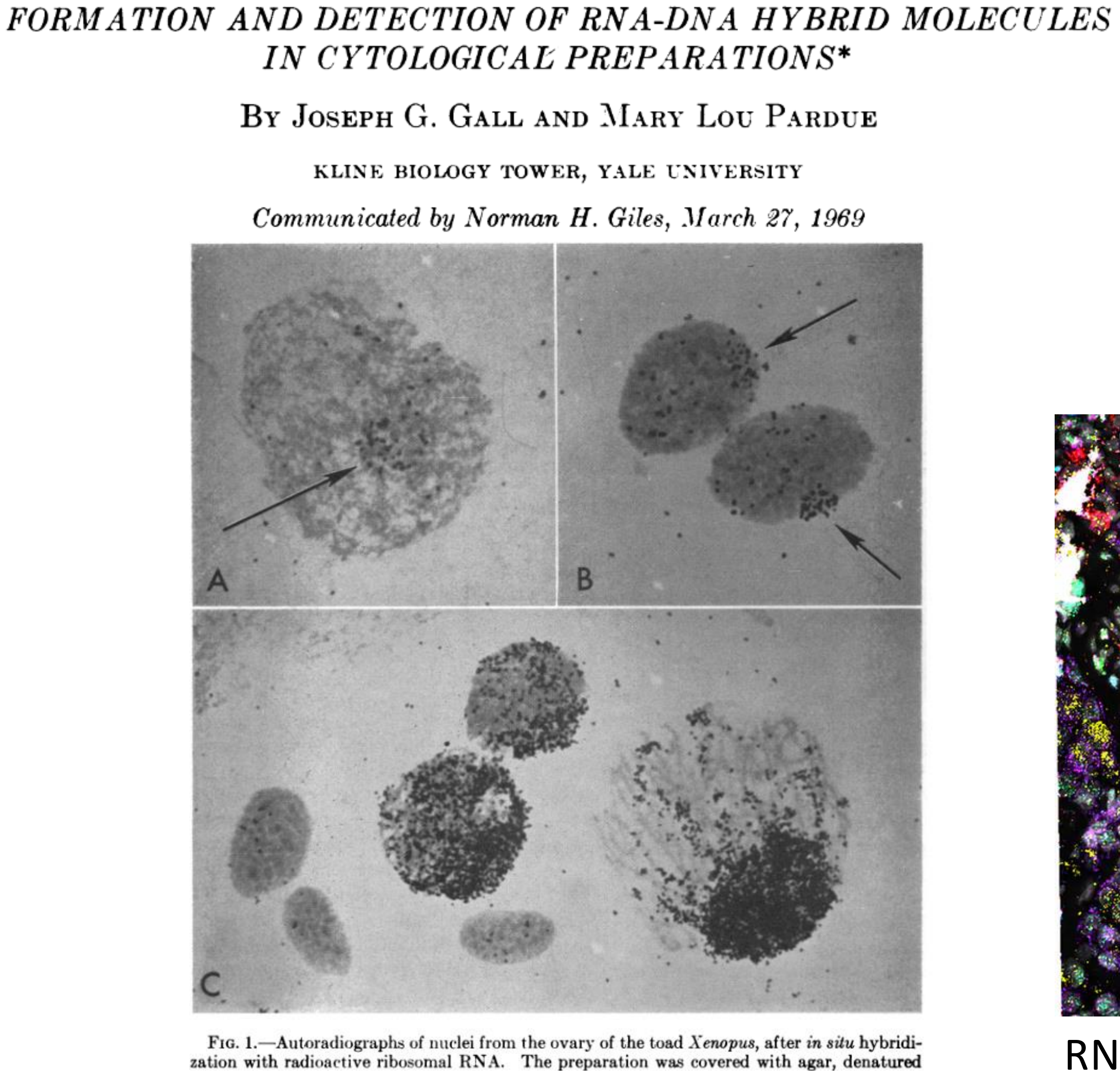
Spatial genomics

Credits: Bo Xia (NYU)



Ramón Cajal’s depiction of the retina [1904]
Cajal Institute, Spanish National Research Council

Spatial mapping using *in situ* hybridization (ISH)



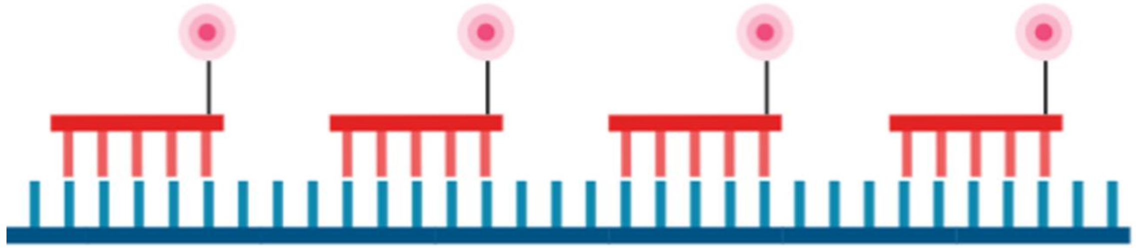
*FORMATION AND DETECTION OF RNA-DNA HYBRID MOLECULES IN CYTOLOGICAL PREPARATIONS**

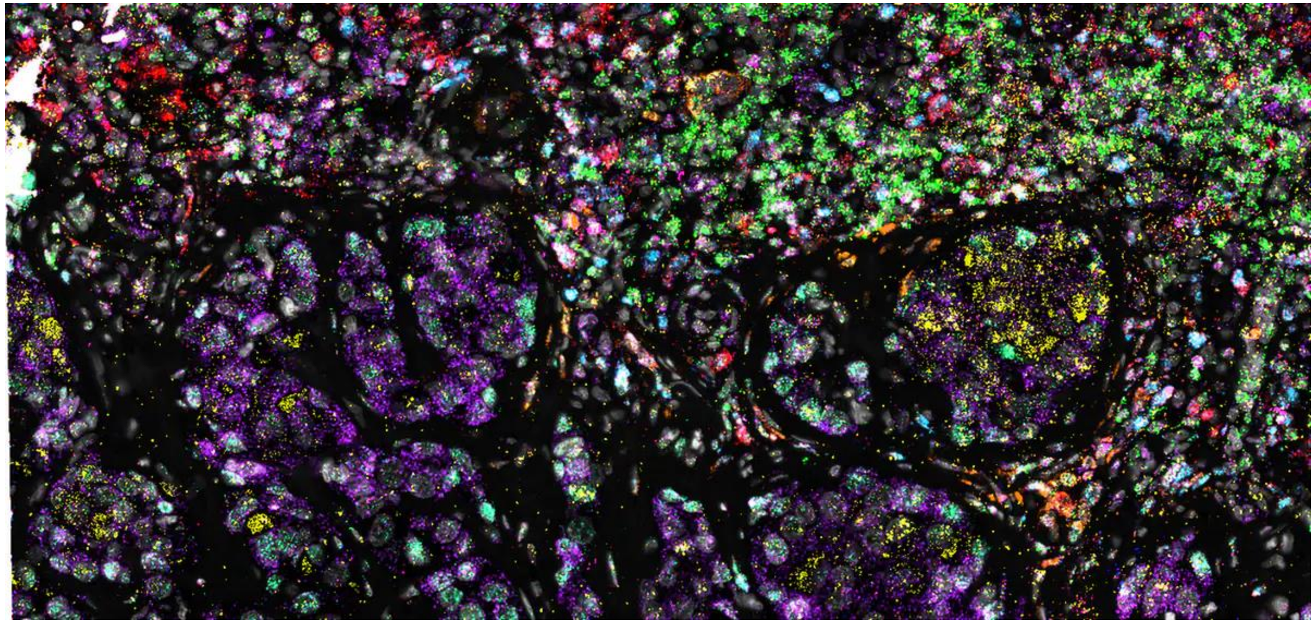
BY JOSEPH G. GALL AND MARY LOU PARDUE

KLINE BIOLOGY TOWER, YALE UNIVERSITY

Communicated by Norman H. Giles, March 27, 1969

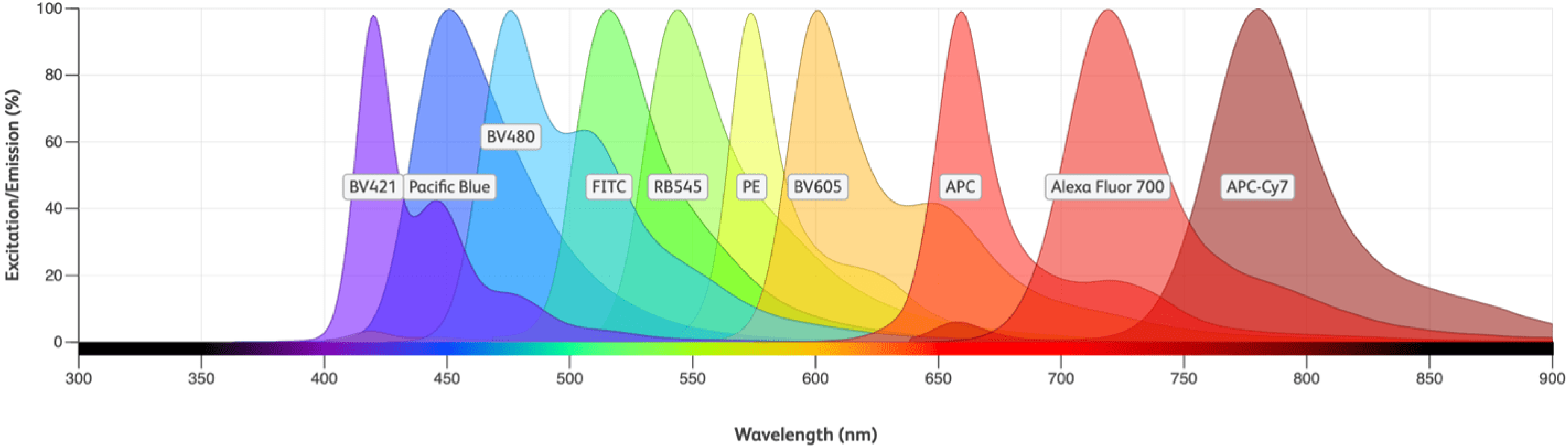
FIG. 1.—Autoradiographs of nuclei from the ovary of the toad *Xenopus*, after *in situ* hybridization with radioactive ribosomal RNA. The preparation was covered with agar, denatured



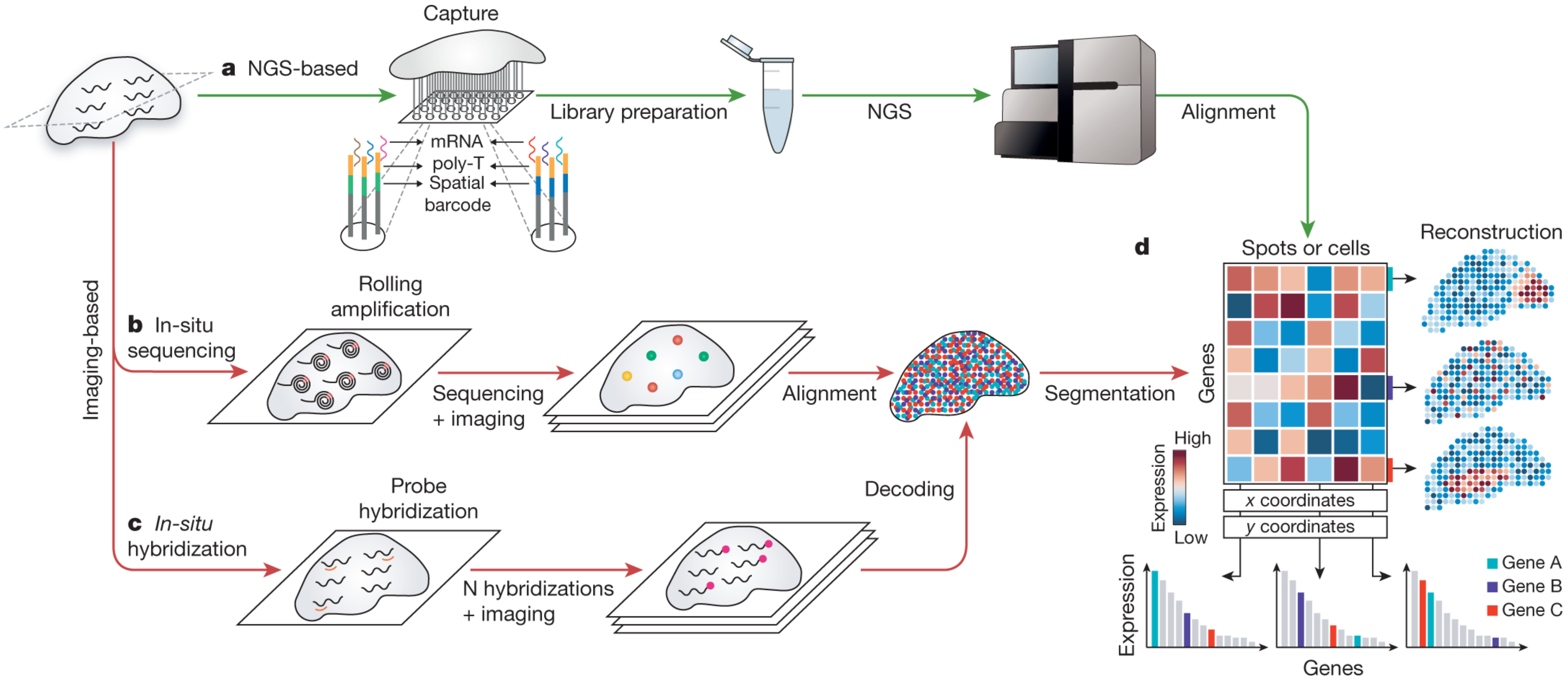


RNAscope (12 RNAs), human lung cancer

How can we scale these spatial mapping techniques to thousands of genes?

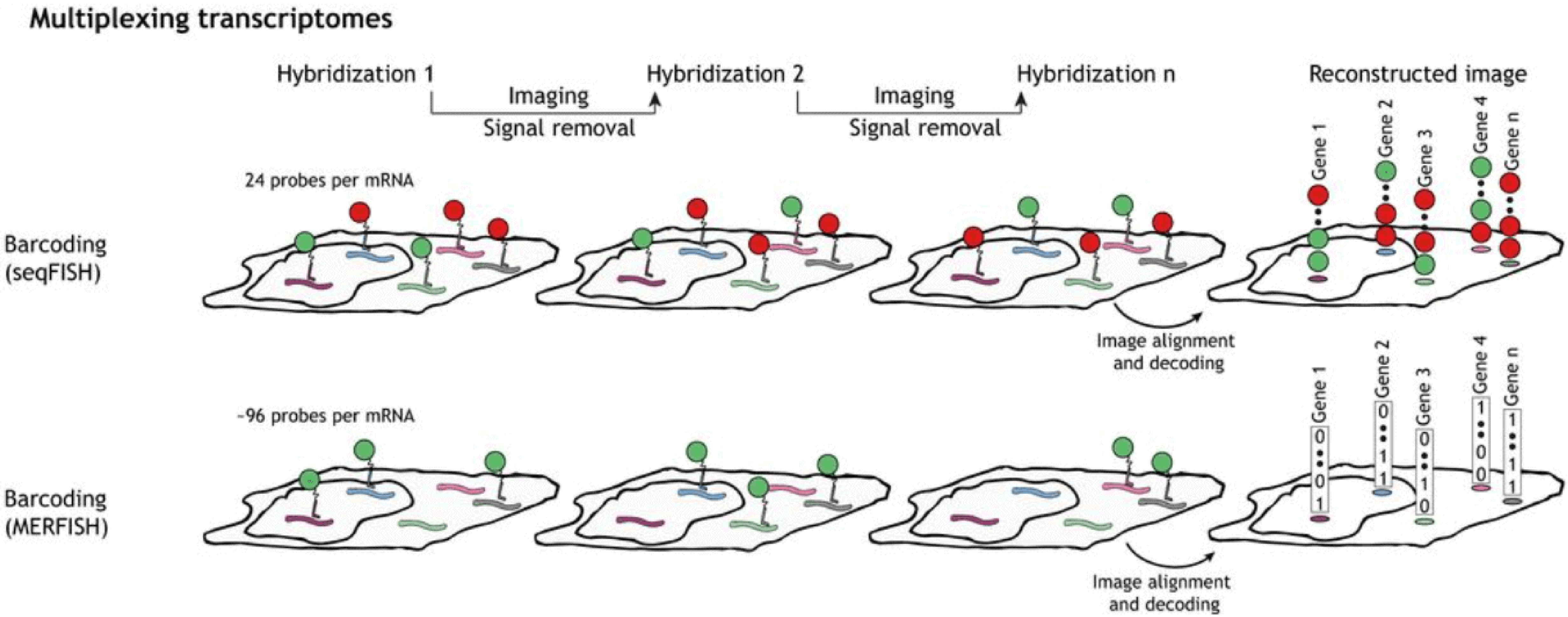


Spatial transcriptomics technologies



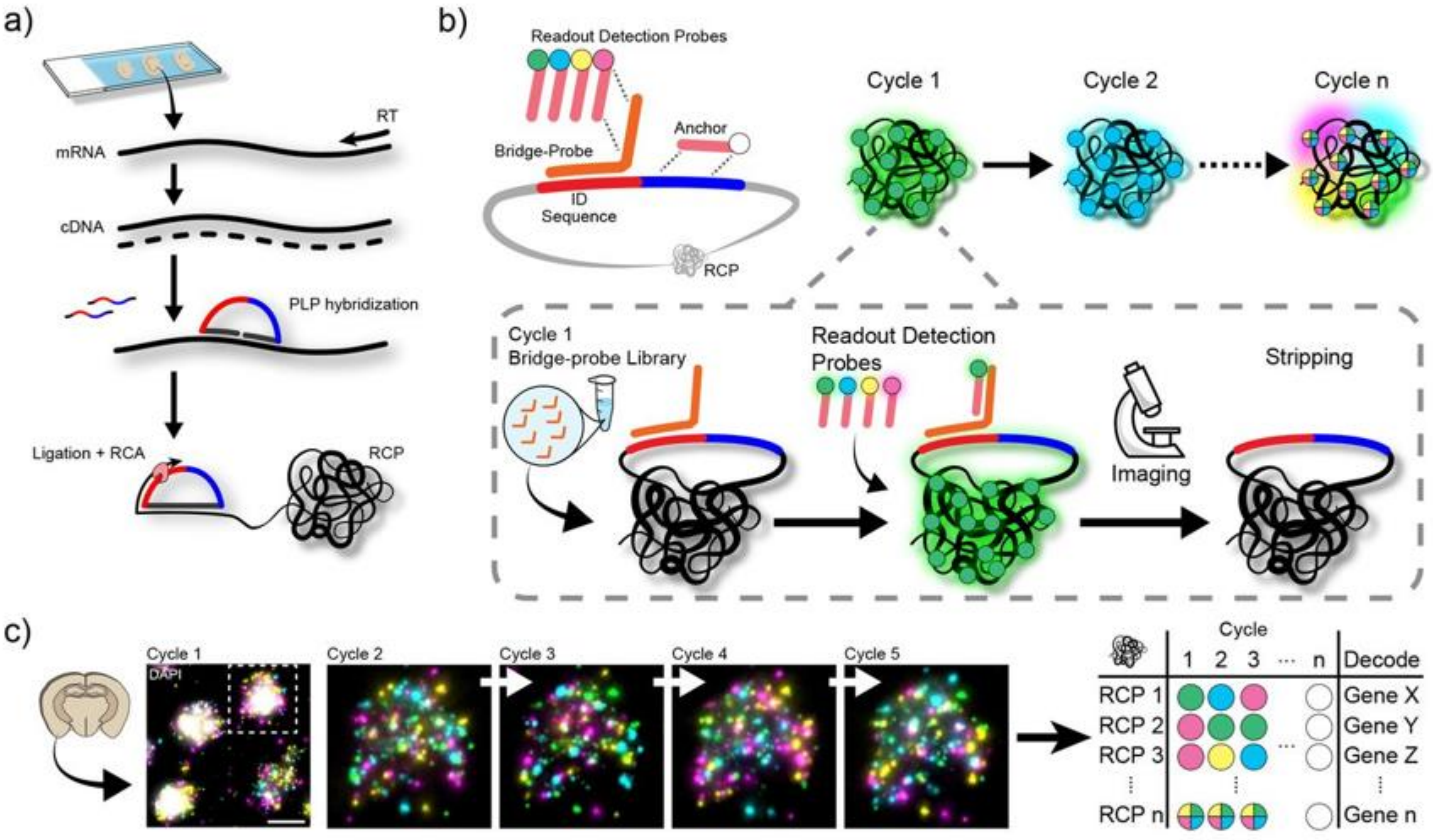
Rao,…, Yanai (2021)

Imaging-based spatial transcriptomics



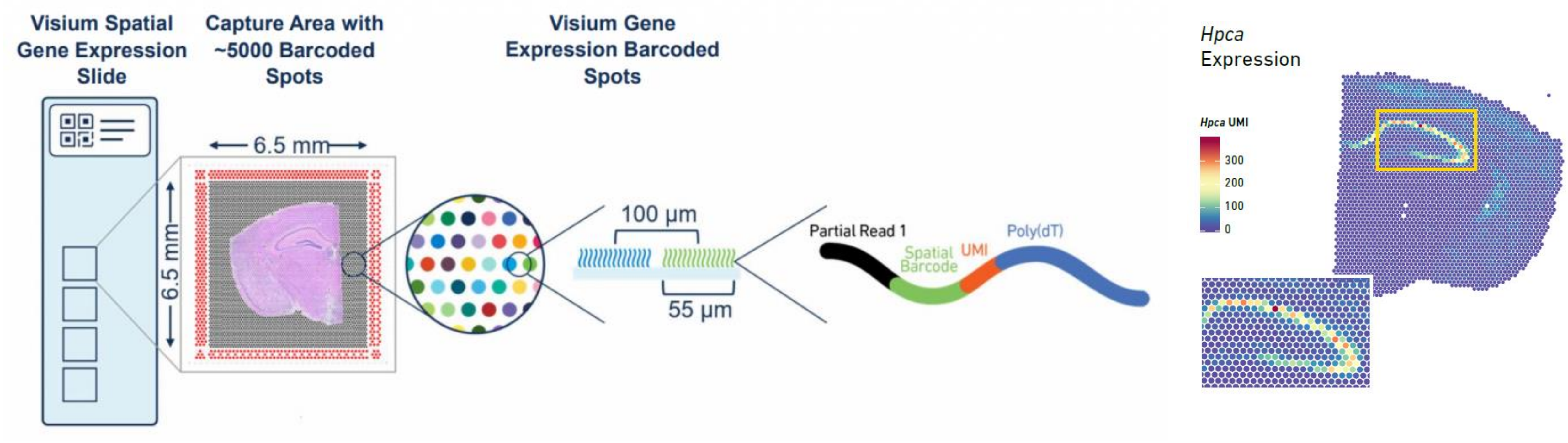
Mayr *et al.* Development, 2019

In situ sequencing

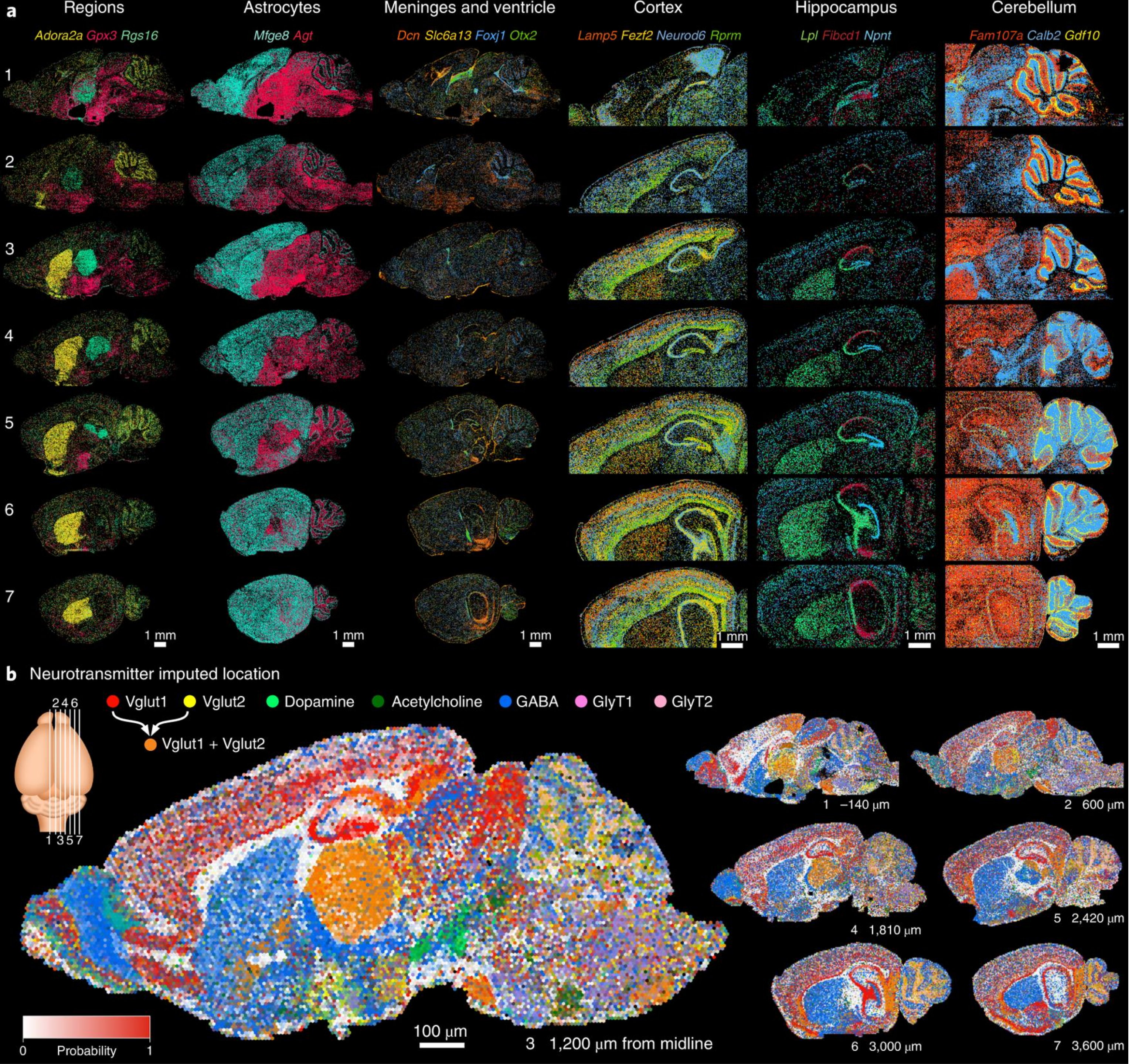


Gyllborg *et al*, Nucleic Acids Research (2020)

Sequencing-based spatial transcriptomics

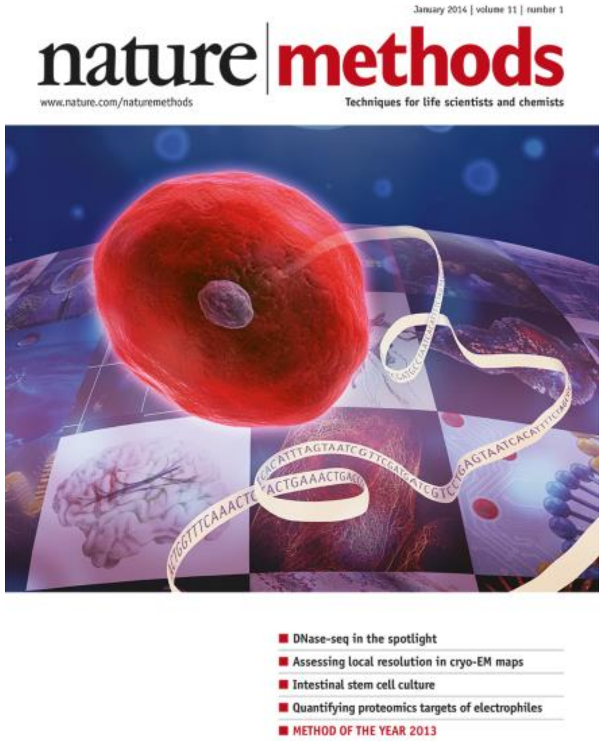


https://www.10xgenomics.com/



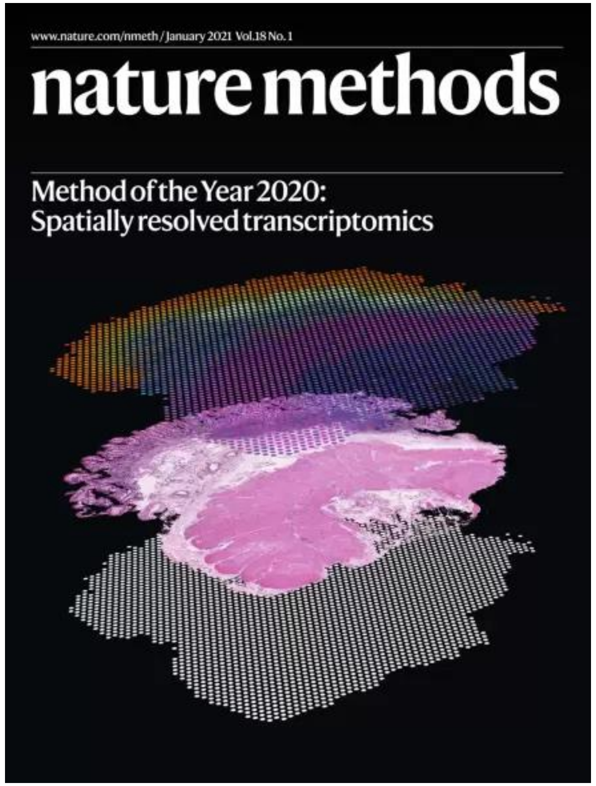
Mouse brain atlas of seven sagittal sections (EELFISH, 168 genes).
Borm et al. Nature Biotechnology 2022

2013
Single-cell sequencing

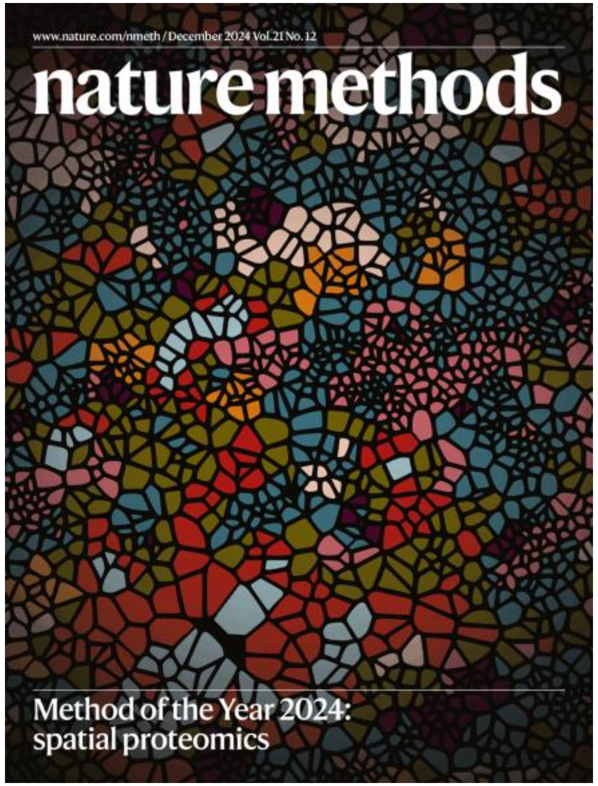


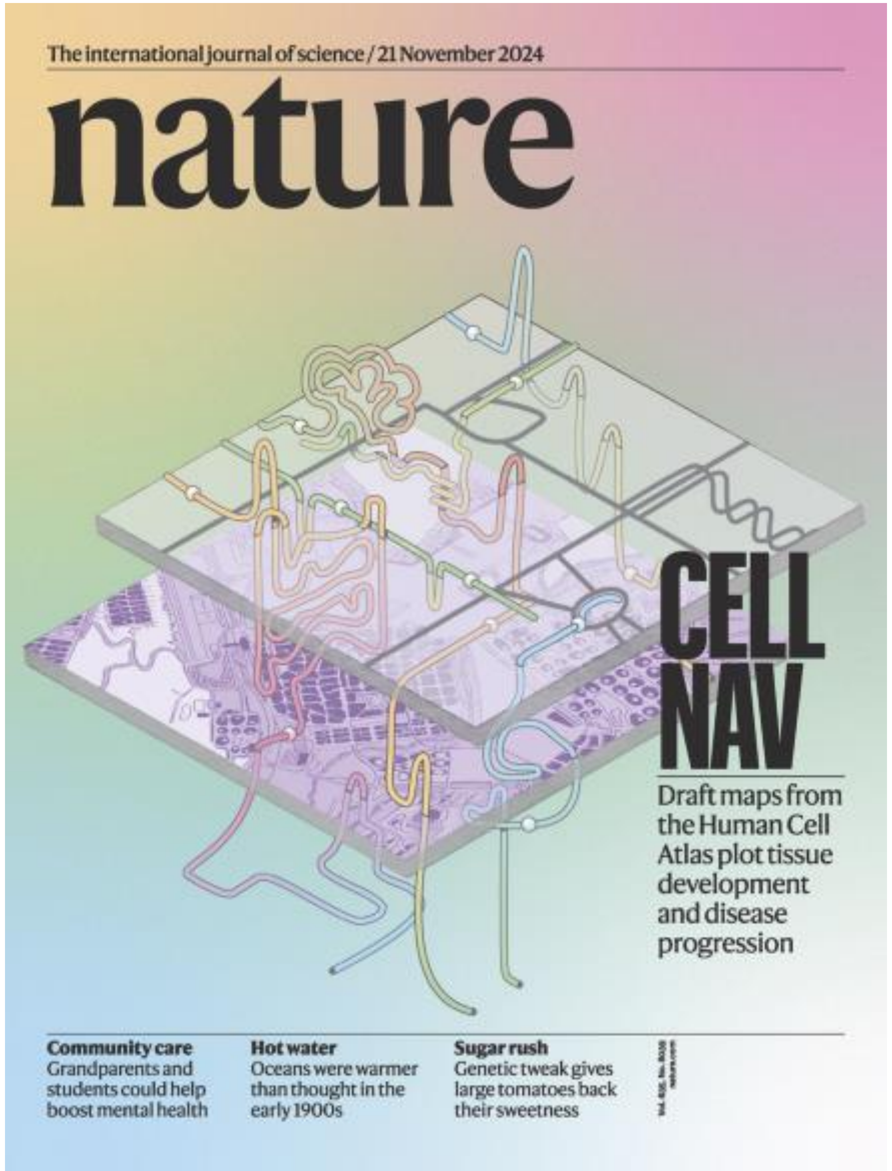
2019
Single-cell multimodal omics

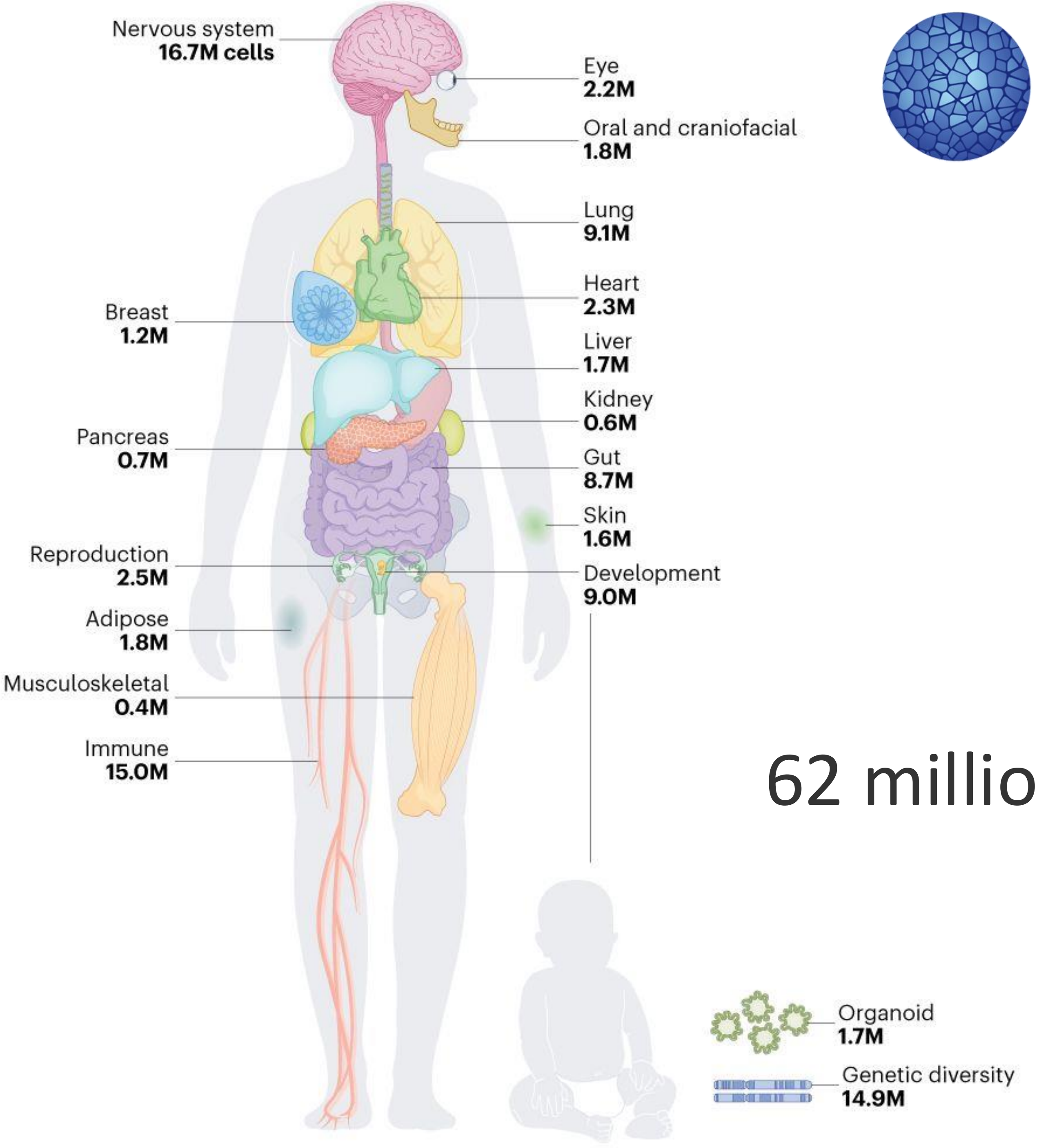
2020
Spatial transcriptomics



2024
Spatial proteomics







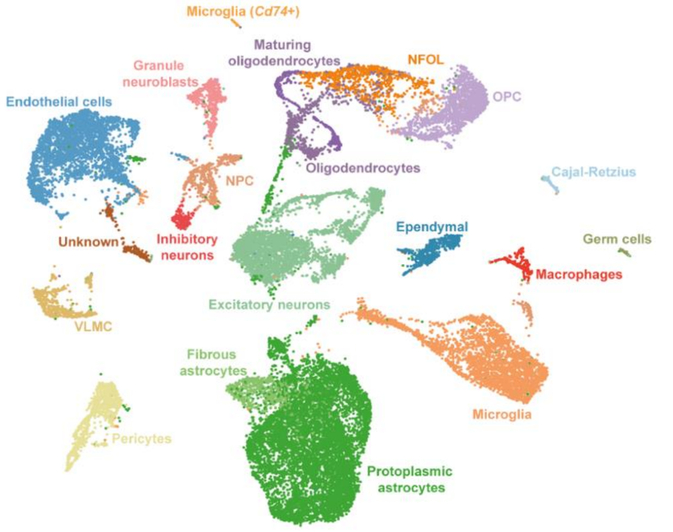
HUMAN CELL ATLAS

62 million cells

https://www.nature.com/immersive/d42859-024-00060-5

Different technologies answer different questions

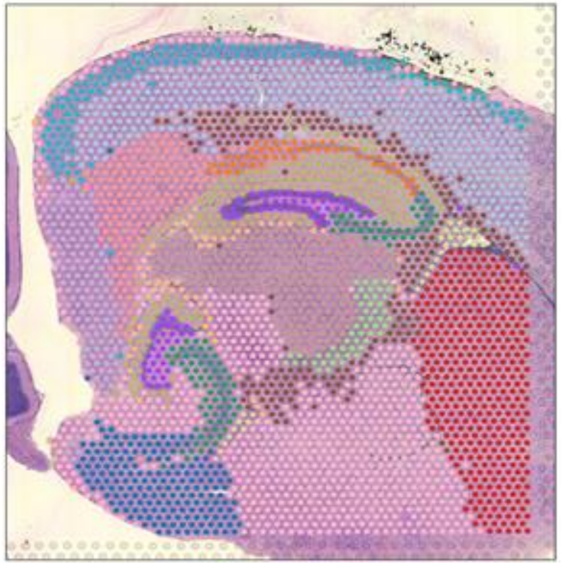
Single-cell/nuclei RNA-seq



+ "Full transcriptome"

- No spatial information

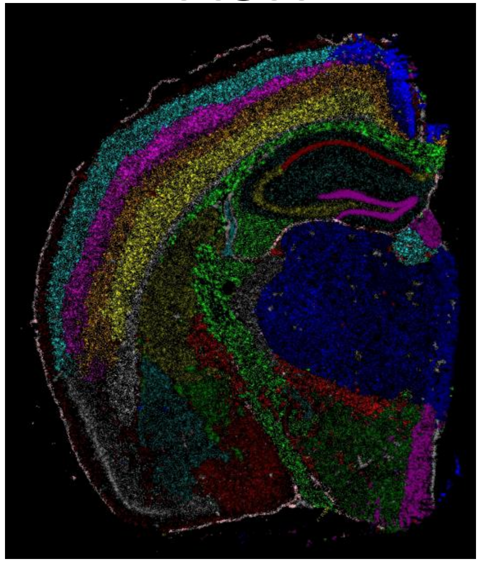
NGS-based



+ "Full transcriptome"

- Low resolution (spots)

In-situ sequencing / x-FISH



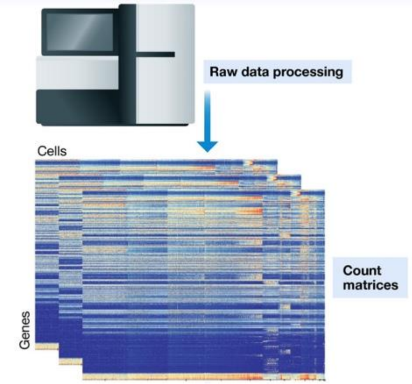
+ High resolution

- Limited genomic features

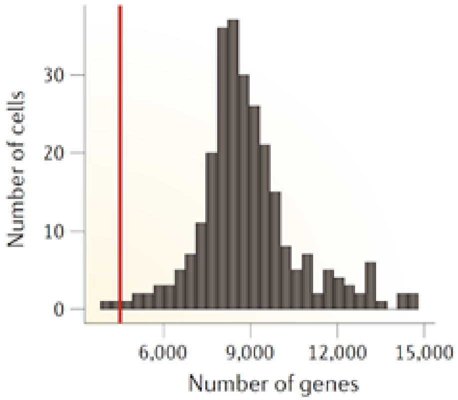
How do we analyze scRNA-seq data?

scRNA-seq analysis workflow

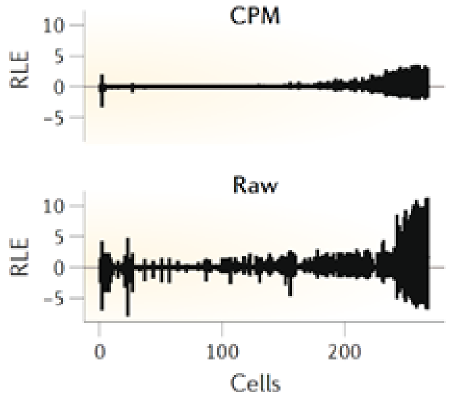
Preprocessing



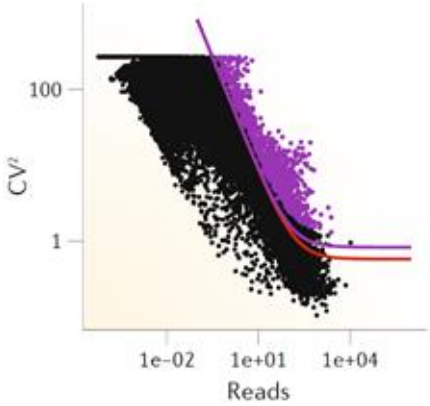
Quality control



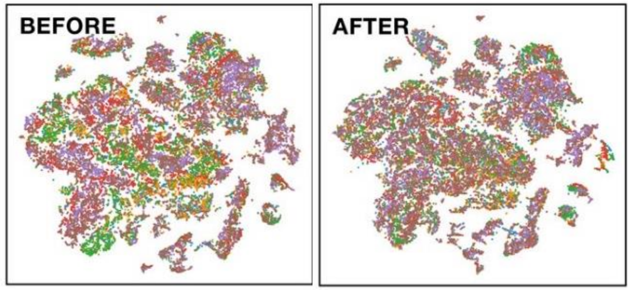
Normalization



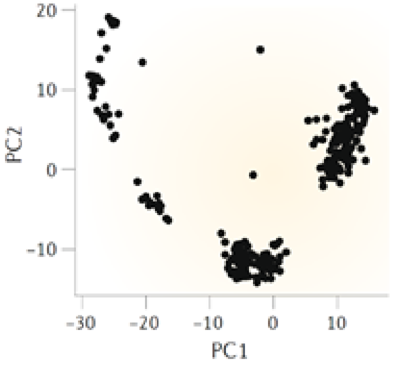
Feature (gene) selection



Batch correction

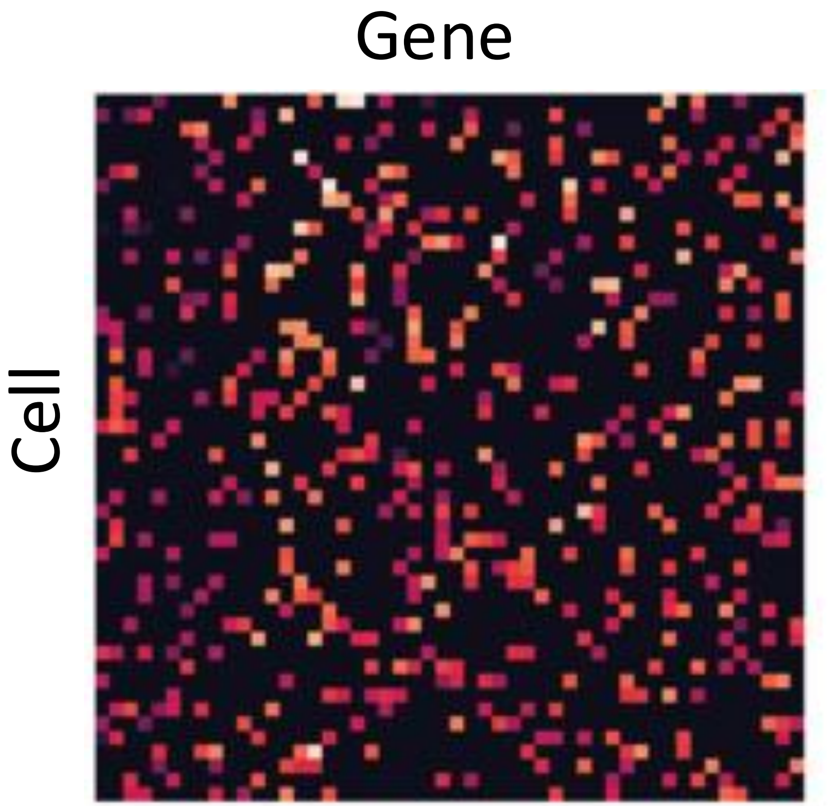


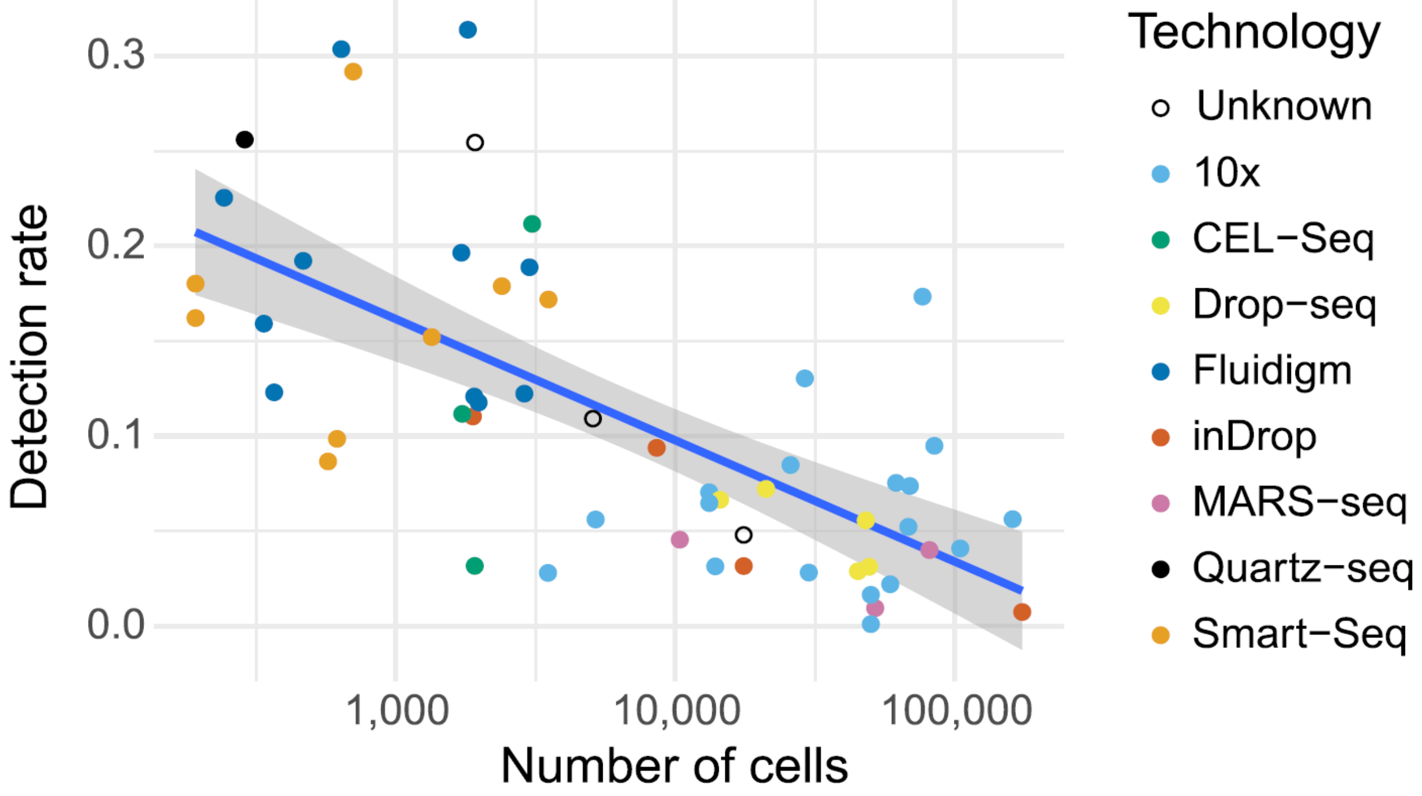
Dimensionality reduction



Luecken and Theis (MSB 2019), Kiselev et al. (Nature review Genetics 2019)

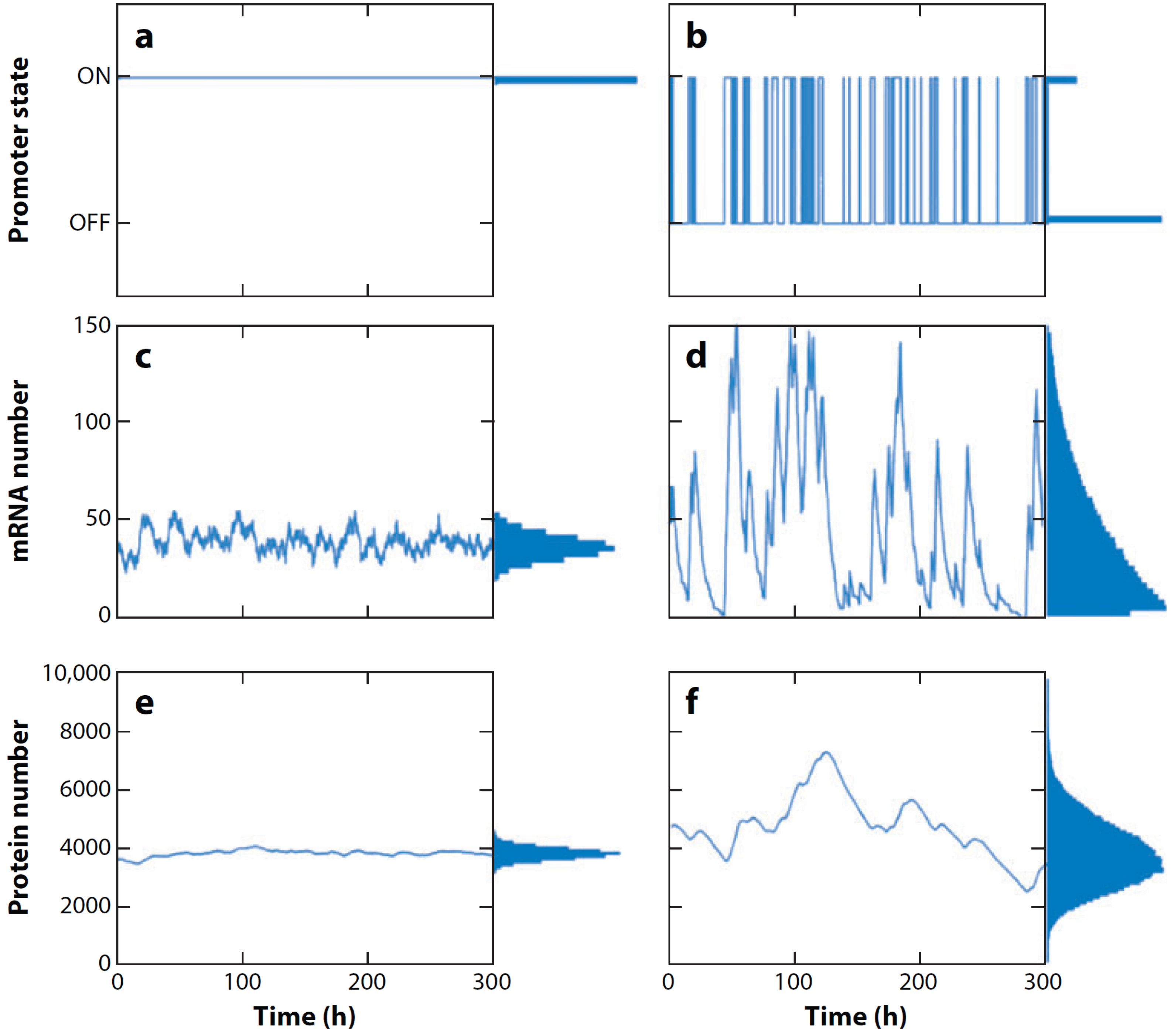
scRNA-seq data is very sparse





Bouland et al. (Genome Biology 2023)

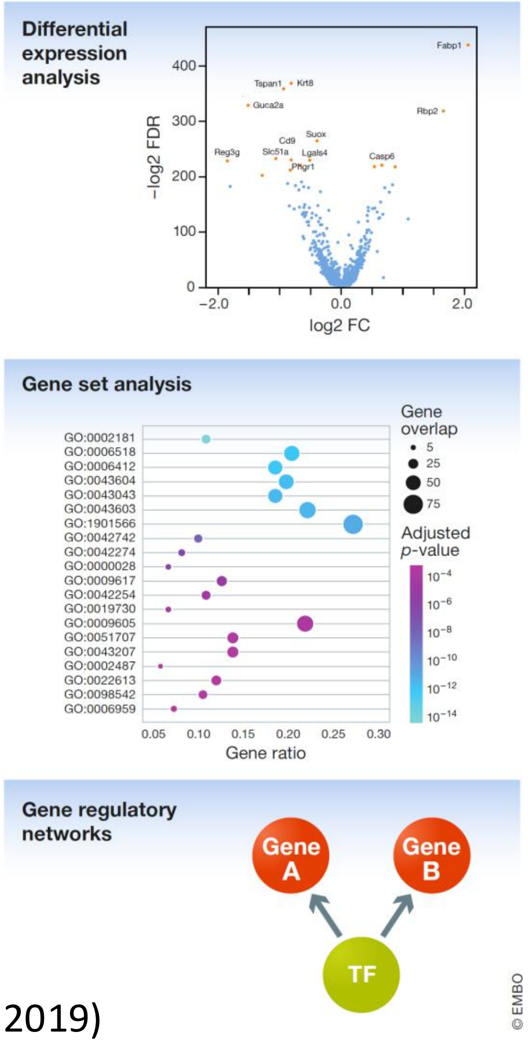
Transcriptional bursting



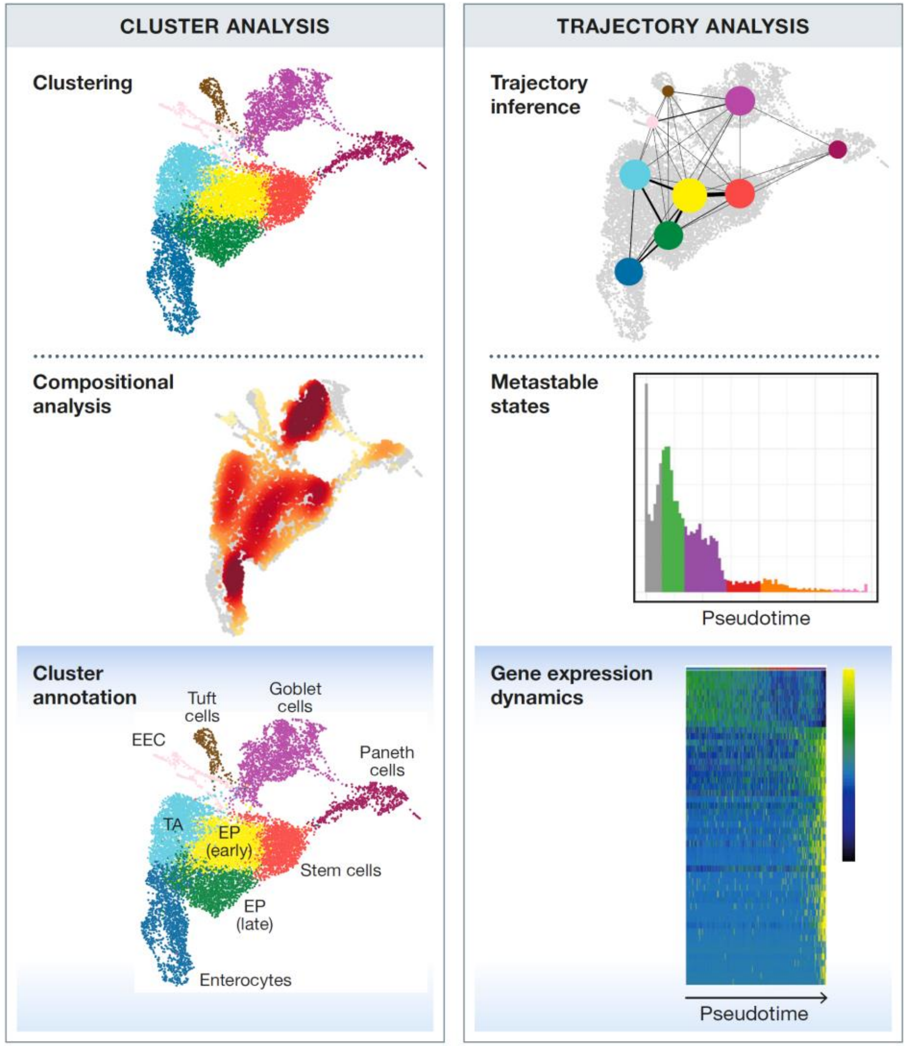
Raj & van Oudenaarden (2009)

What to do with scRNA-seq data?

Gene-level analysis



Cell-level analysis



Luecken and Theis (MSB 2019)

Summary 1

- Single-cell RNA sequencing allows unbiased description of cell states

- Many scRNA-seq protocols available, choice depends on: availability, number of cells, cost, full length vs 3’/5’

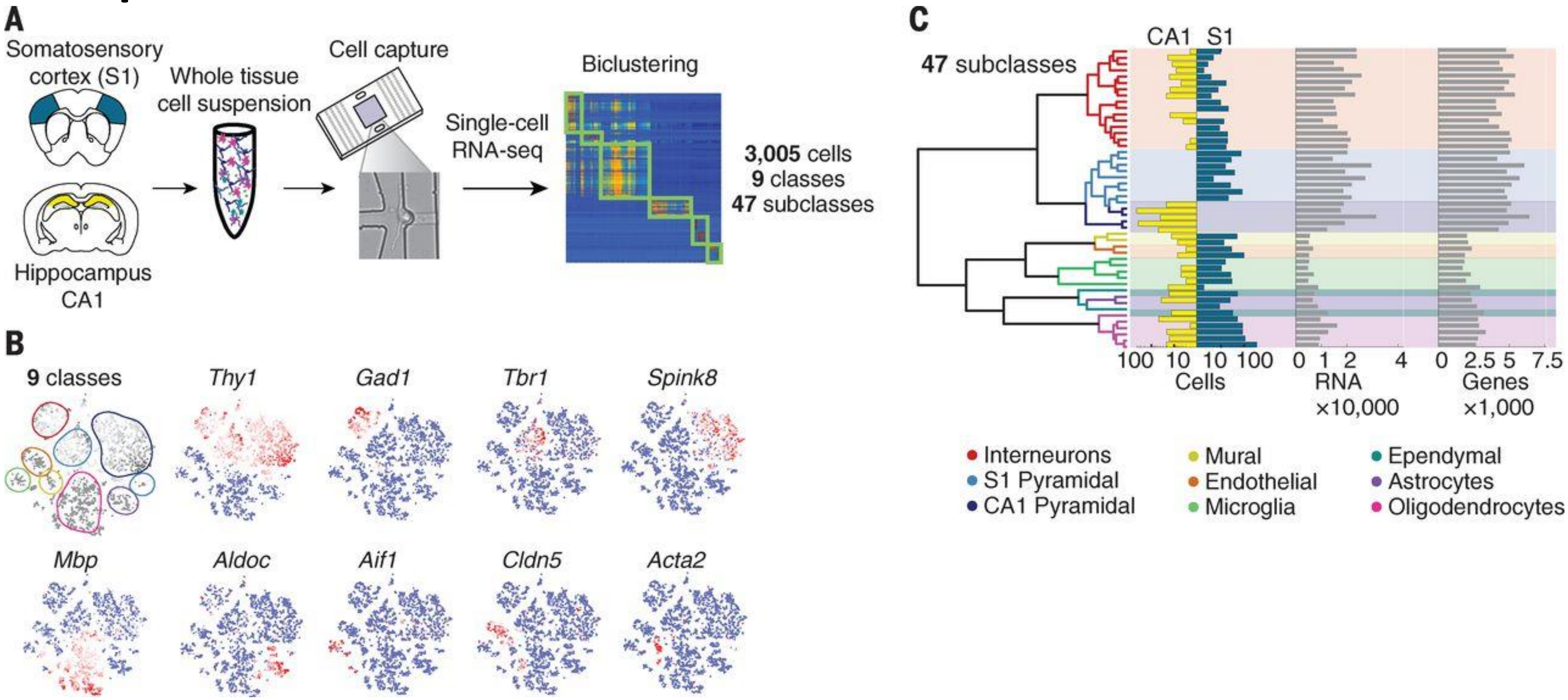
- Several single cell omics measurements, including multi-omics from the same cell

- Spatial transcriptomics allows cell localization

- Computational analysis *is* important

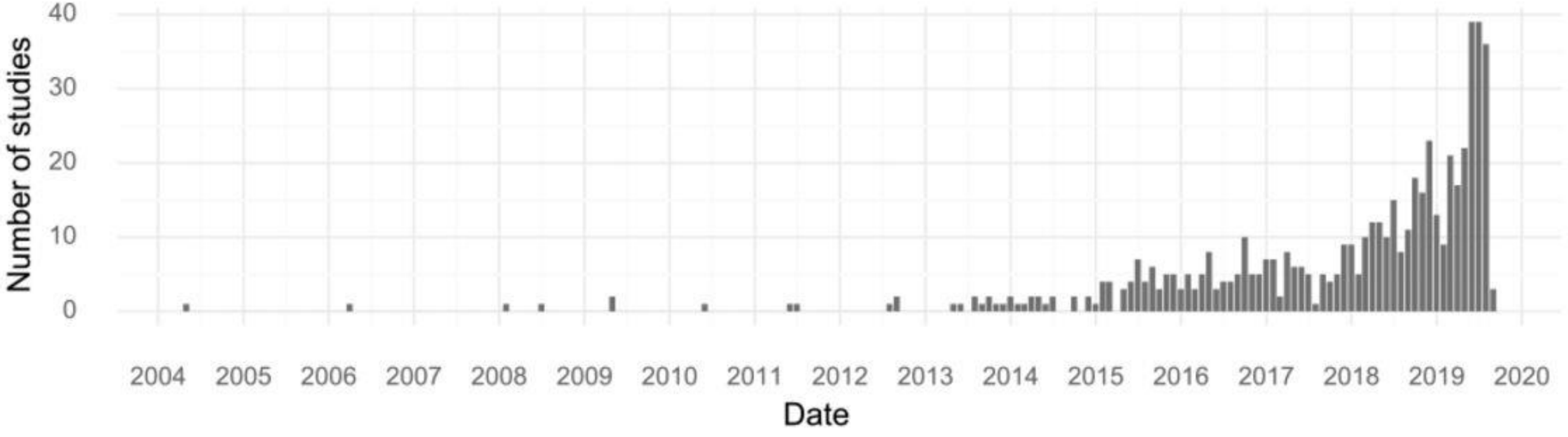
Single-cell RNA-seq in neuroscience

Brain, one of the first organs to be sequenced



Zeisel et al, Science 2015

Brain, the most studied organ

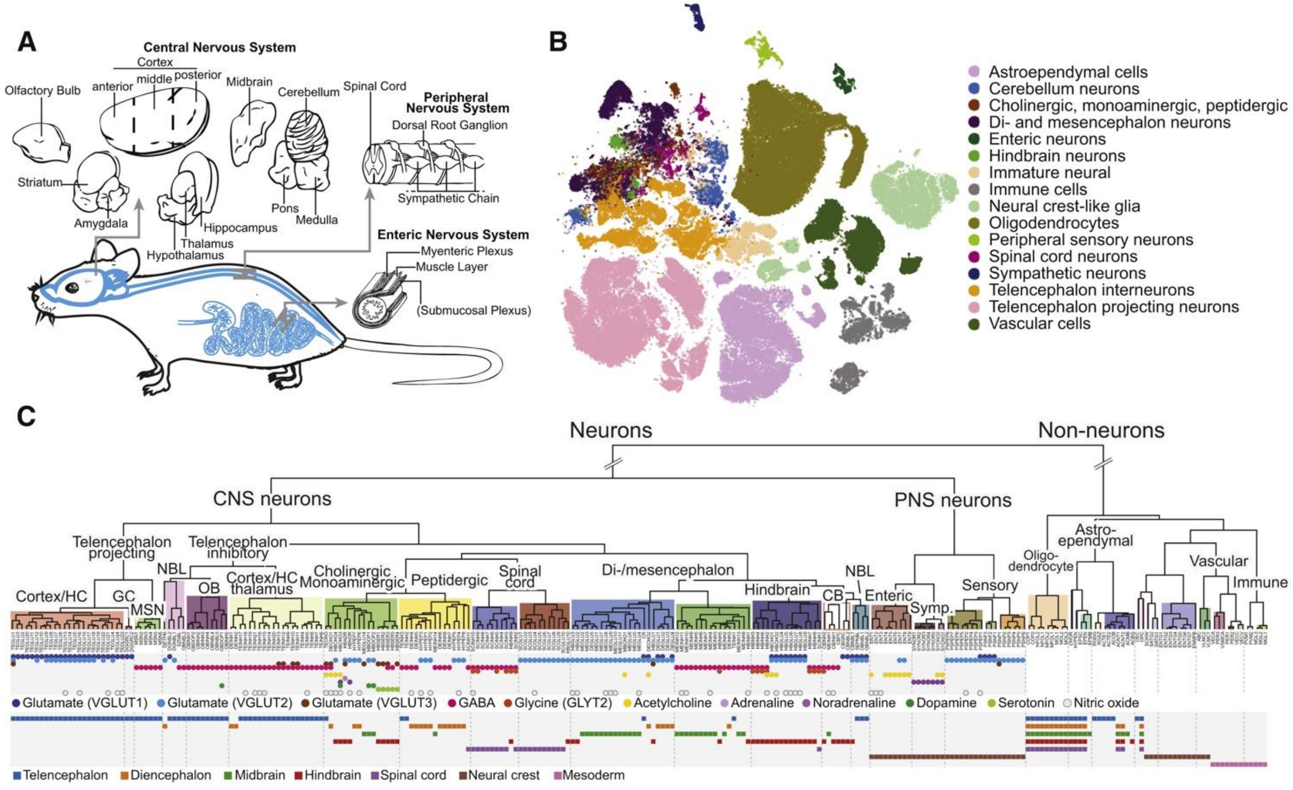


Tissue	Studies
Brain	64
Culture	47
Blood	16
Heart	16
Pancreas	16
Embryo	14
Lung	12

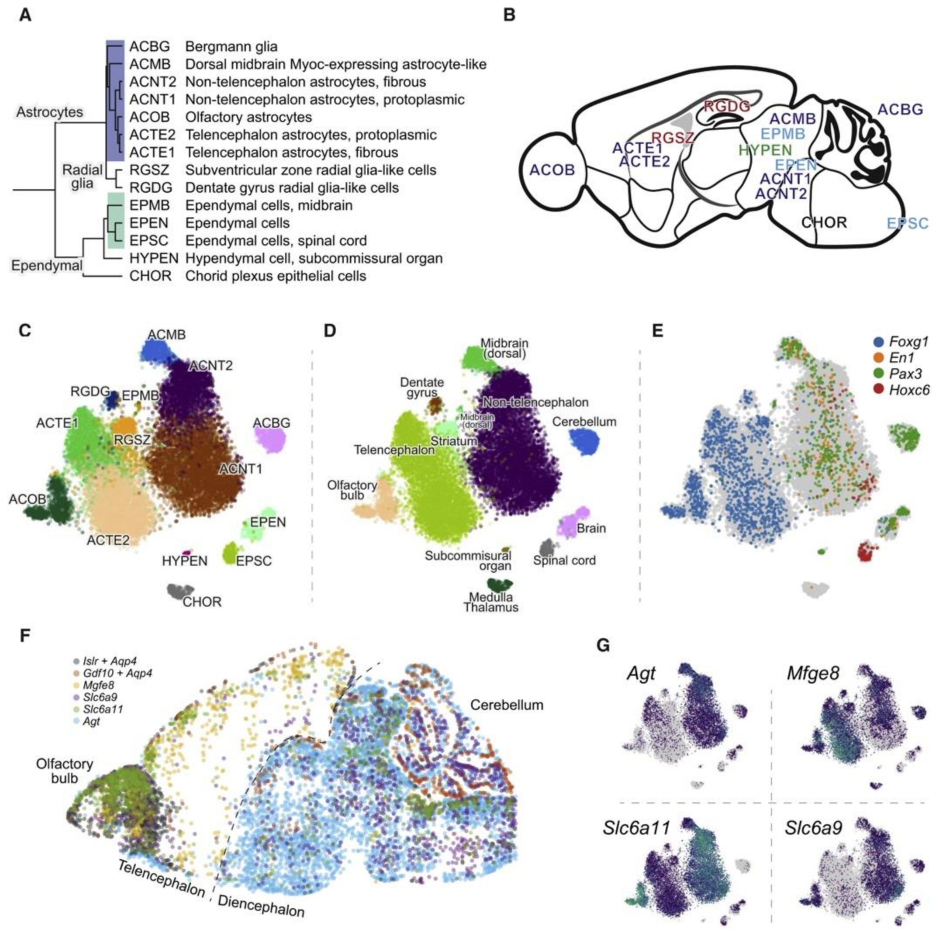
Svensson et al., Database 2020

Build atlases across brain regions: breadth

Adult mouse, 19 regions, ~500k cells



Diversity of the Astroependymal Cells in the CNS

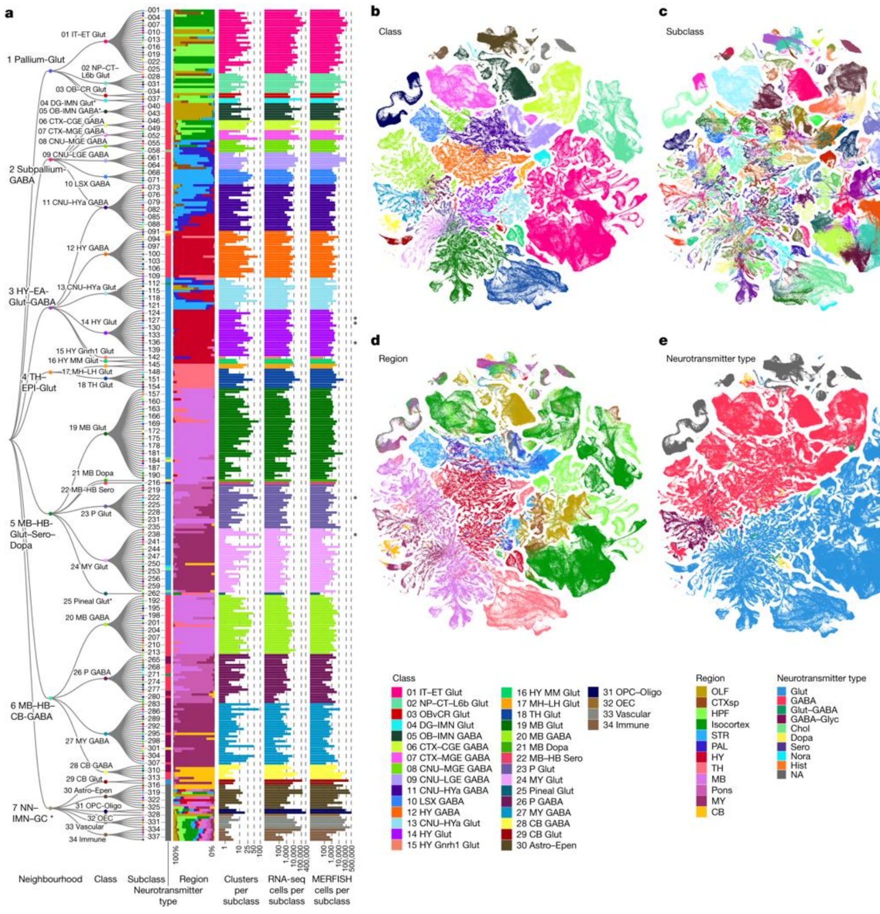


Zeisel et al., *Cell* 2018

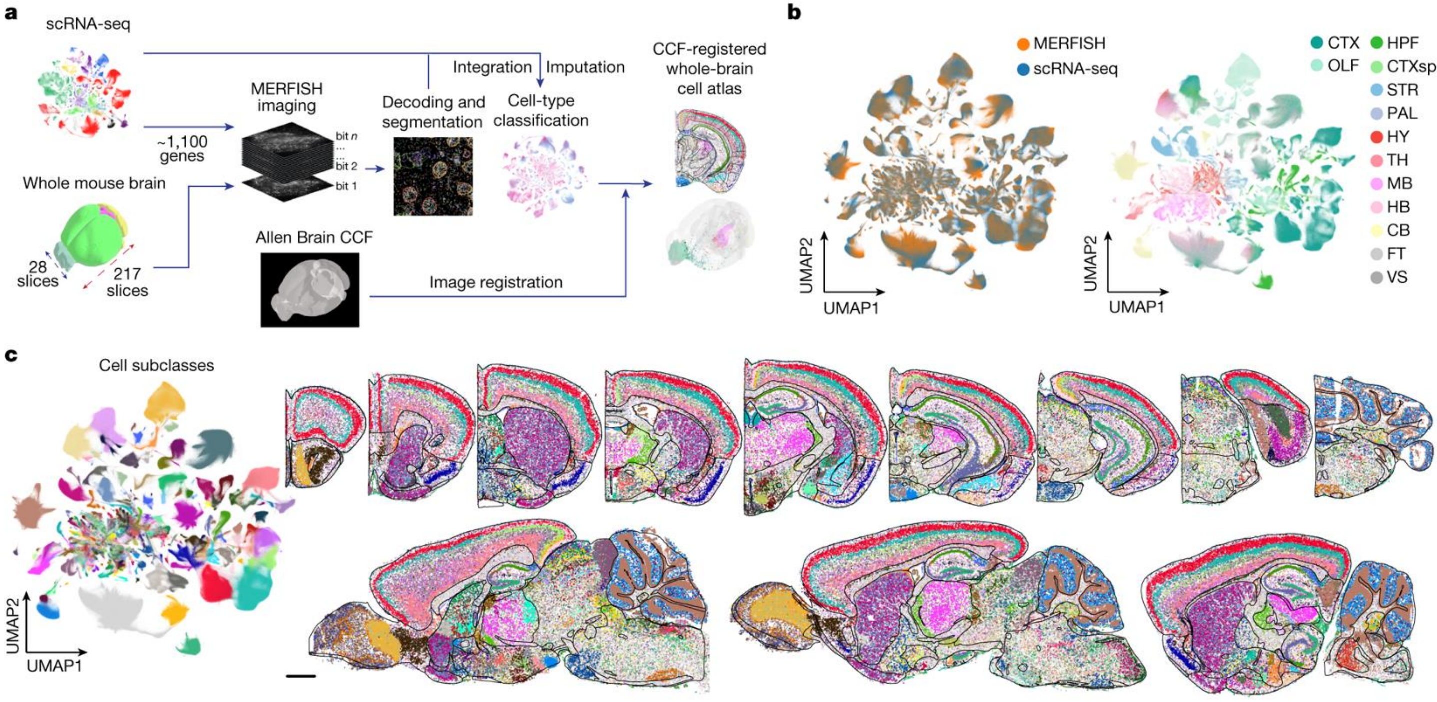
The most comprehensive mouse brain atlases

~4 million cells, whole transcriptome (10x)

~10 million cells, 1100 genes (MERFISH)

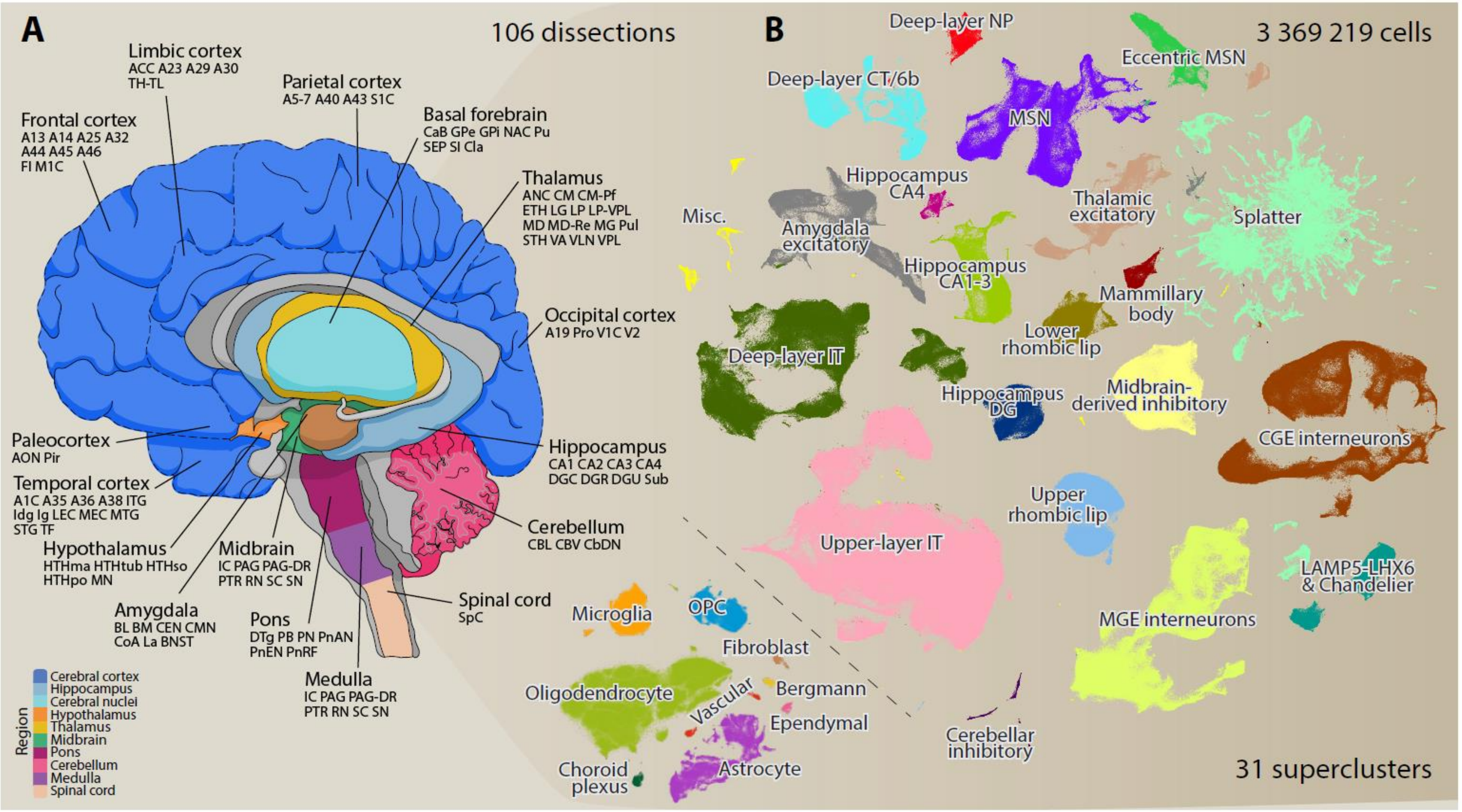


Yao et al., *Nature* 2023



Zhang et al., *Nature* 2023

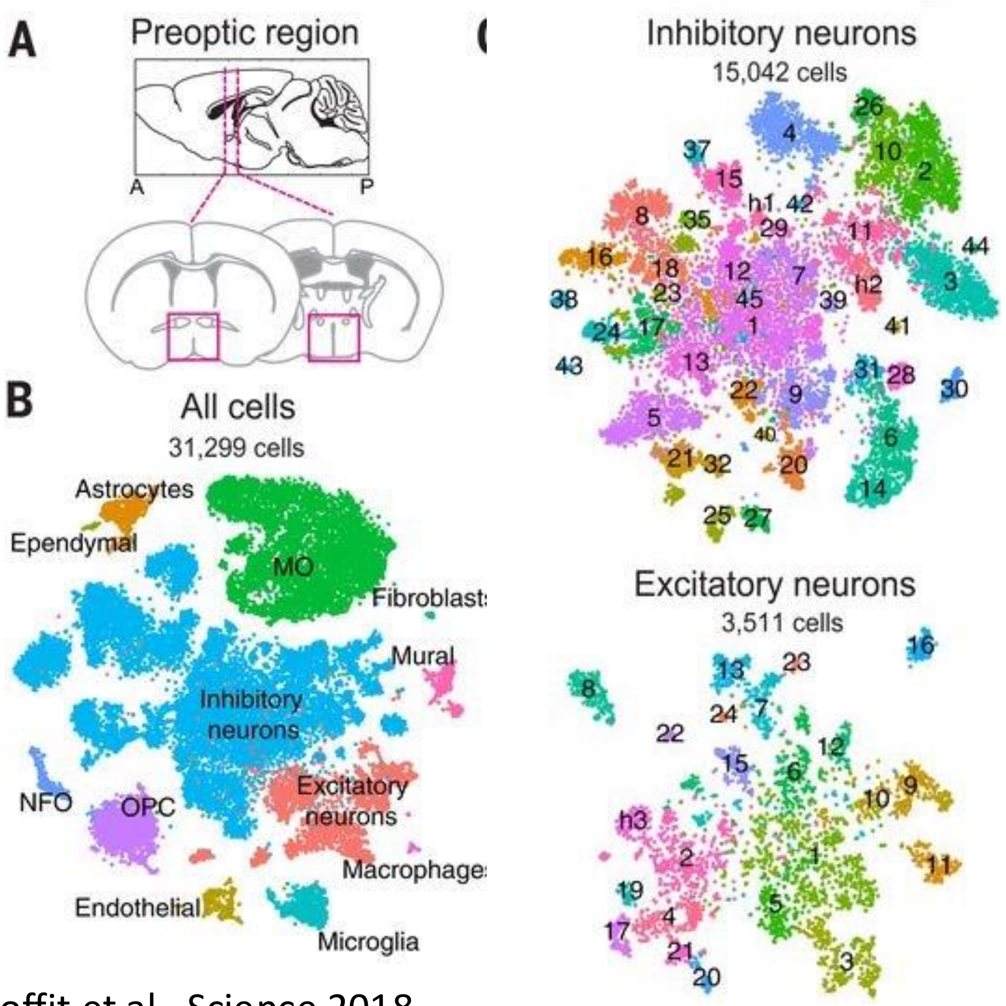
The most comprehensive human brain atlas



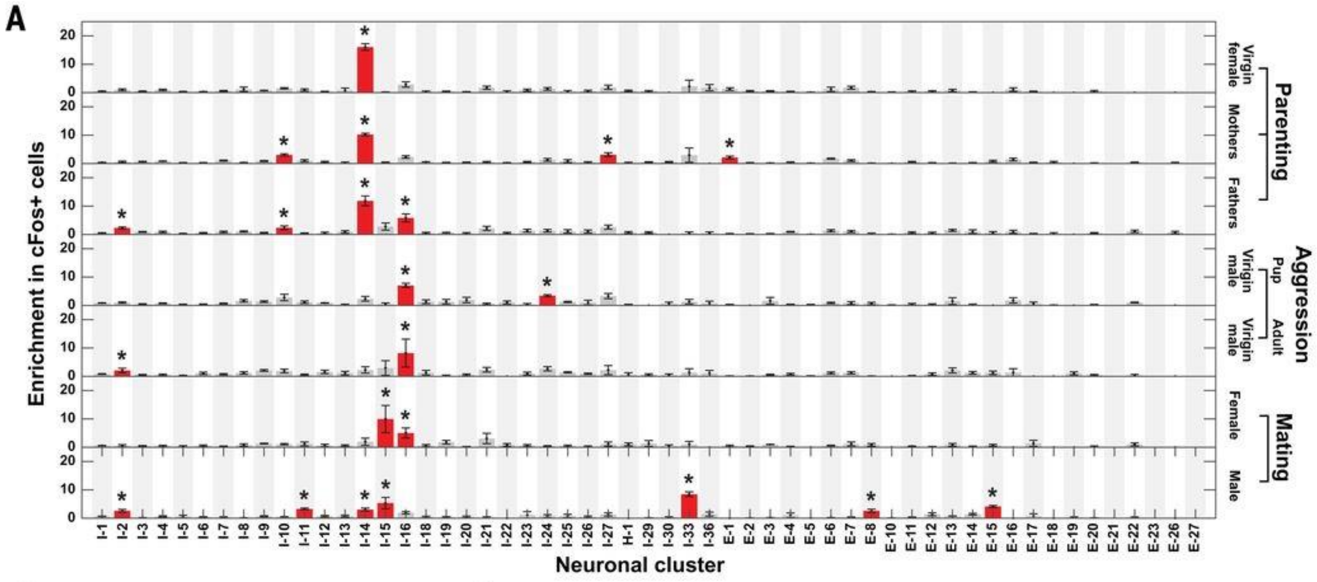
Siletti et al., Science 2023

Build atlases across brain regions: depth

Preoptic region (hypothalamus), ~30k cells



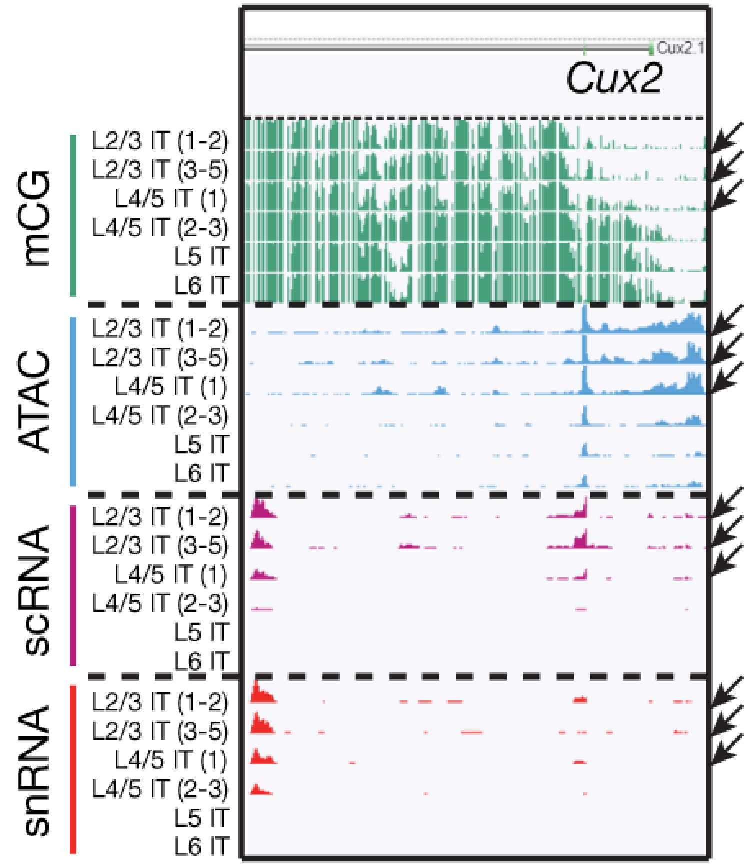
Neuronal clusters activated during specific social behaviors



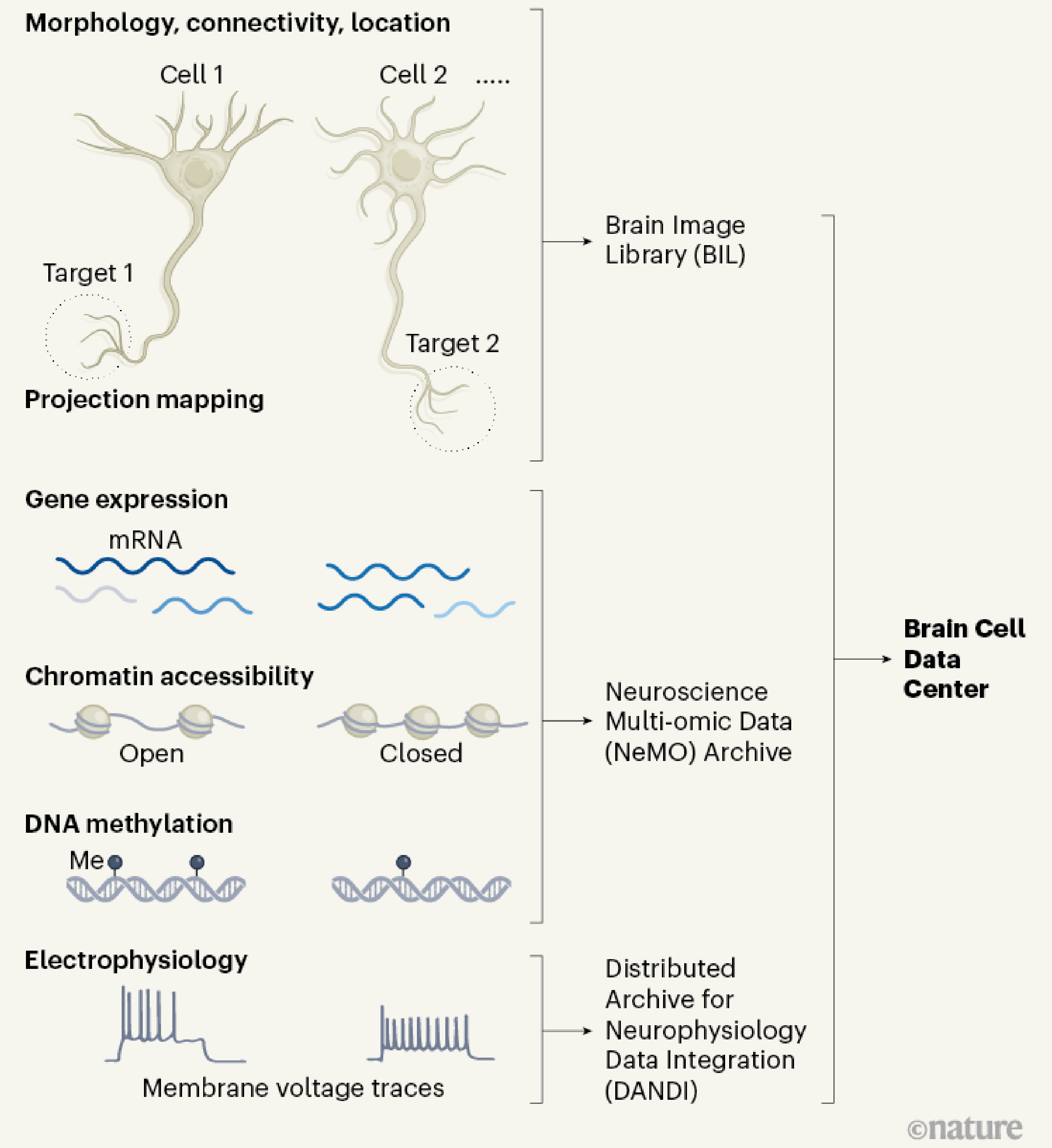
Moffit et al., Science 2018

Multimodal atlases

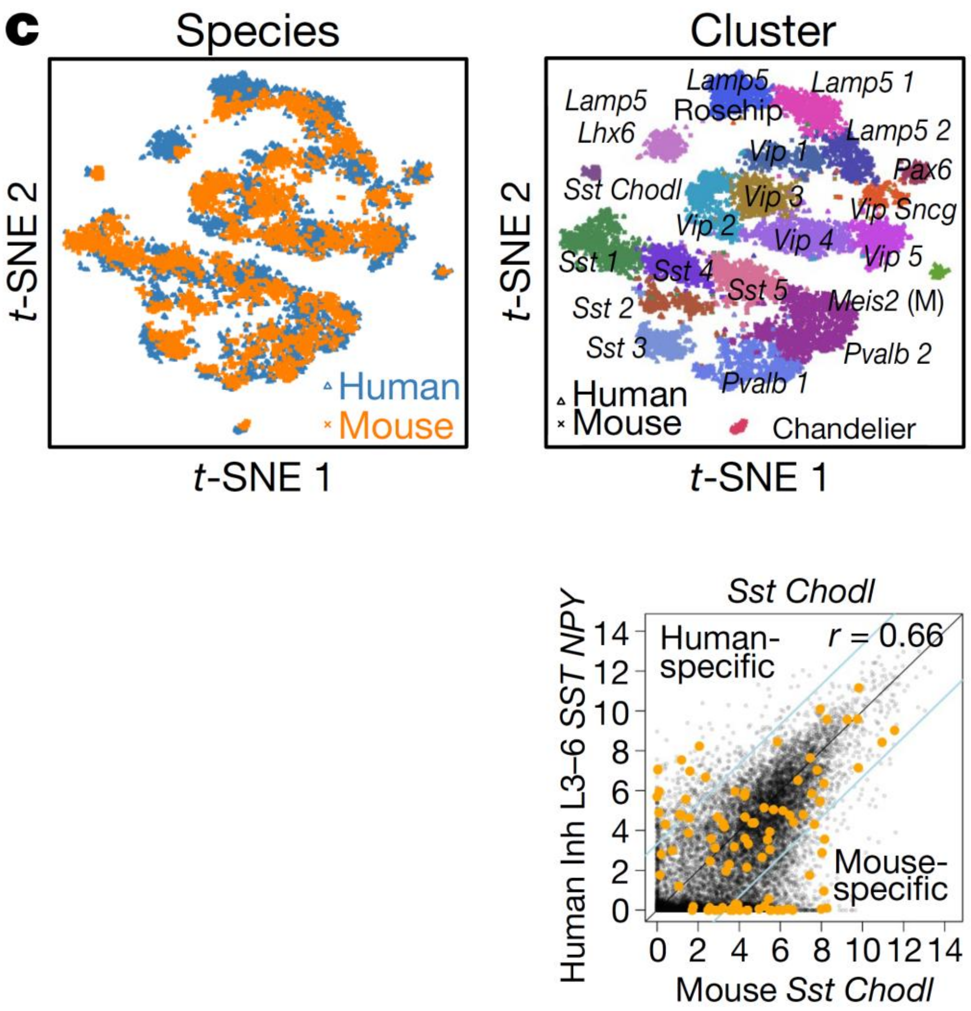
Layer-specific markers

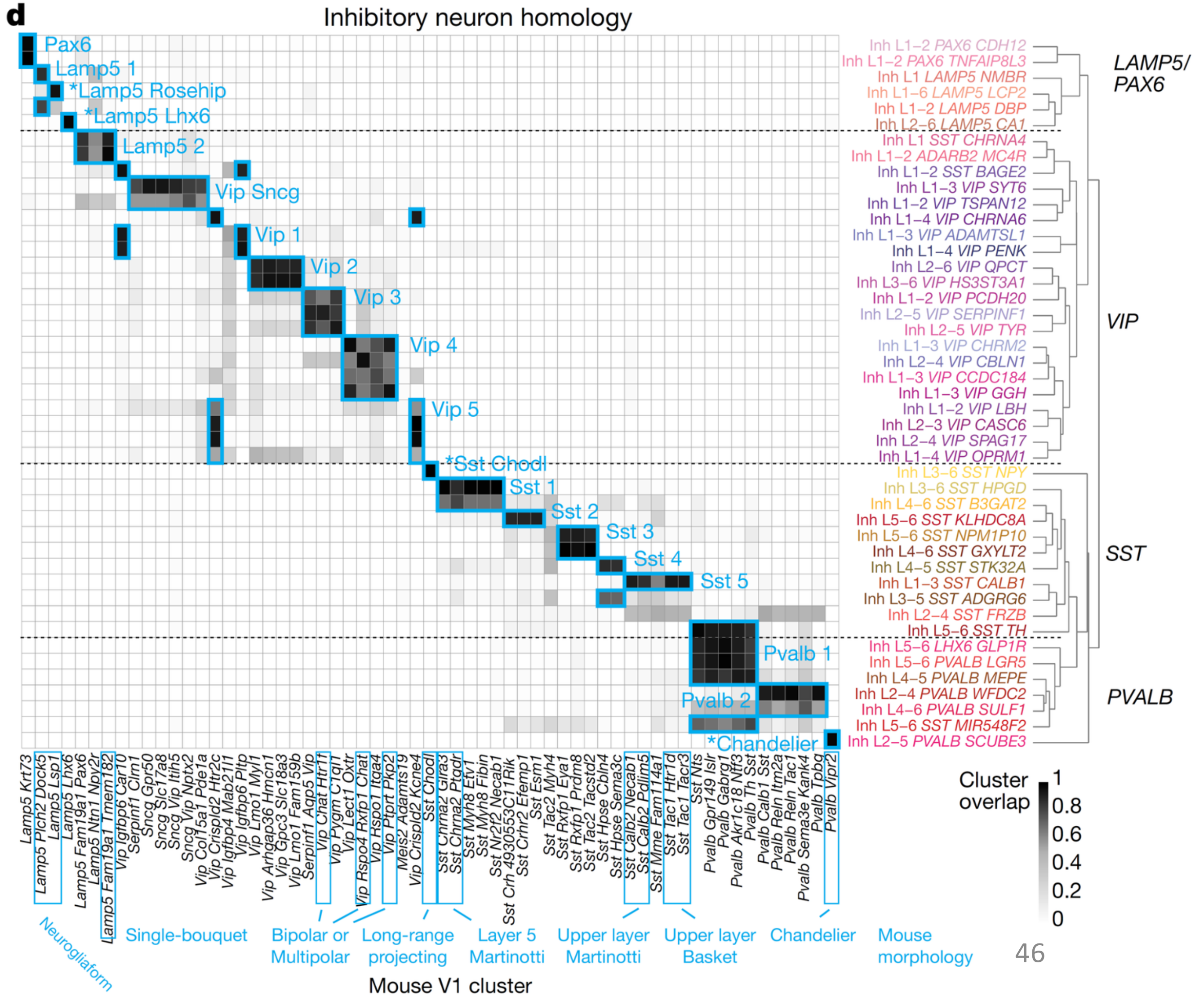


BRAIN Initiative Cell Census Network (BICCN). Nature (2021)



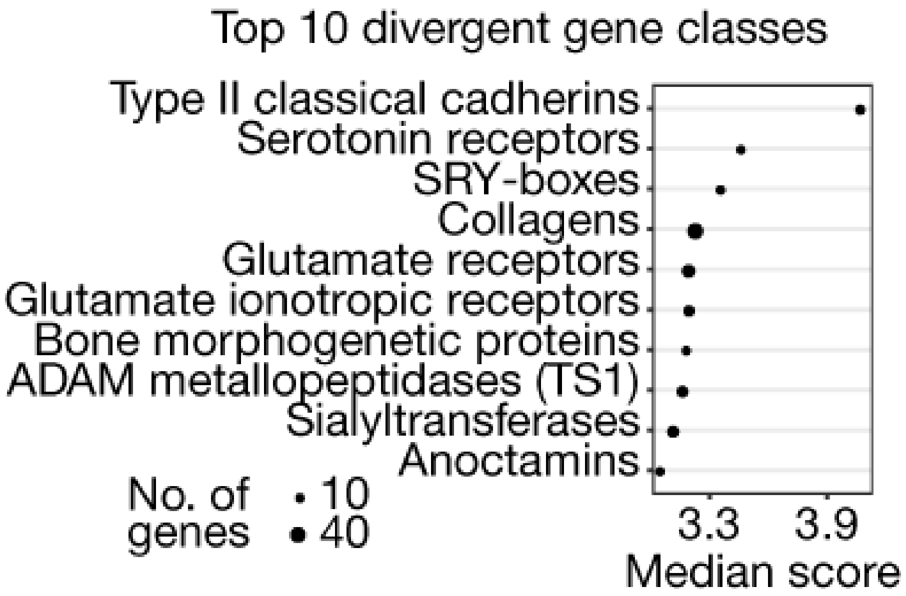
Cell types across species

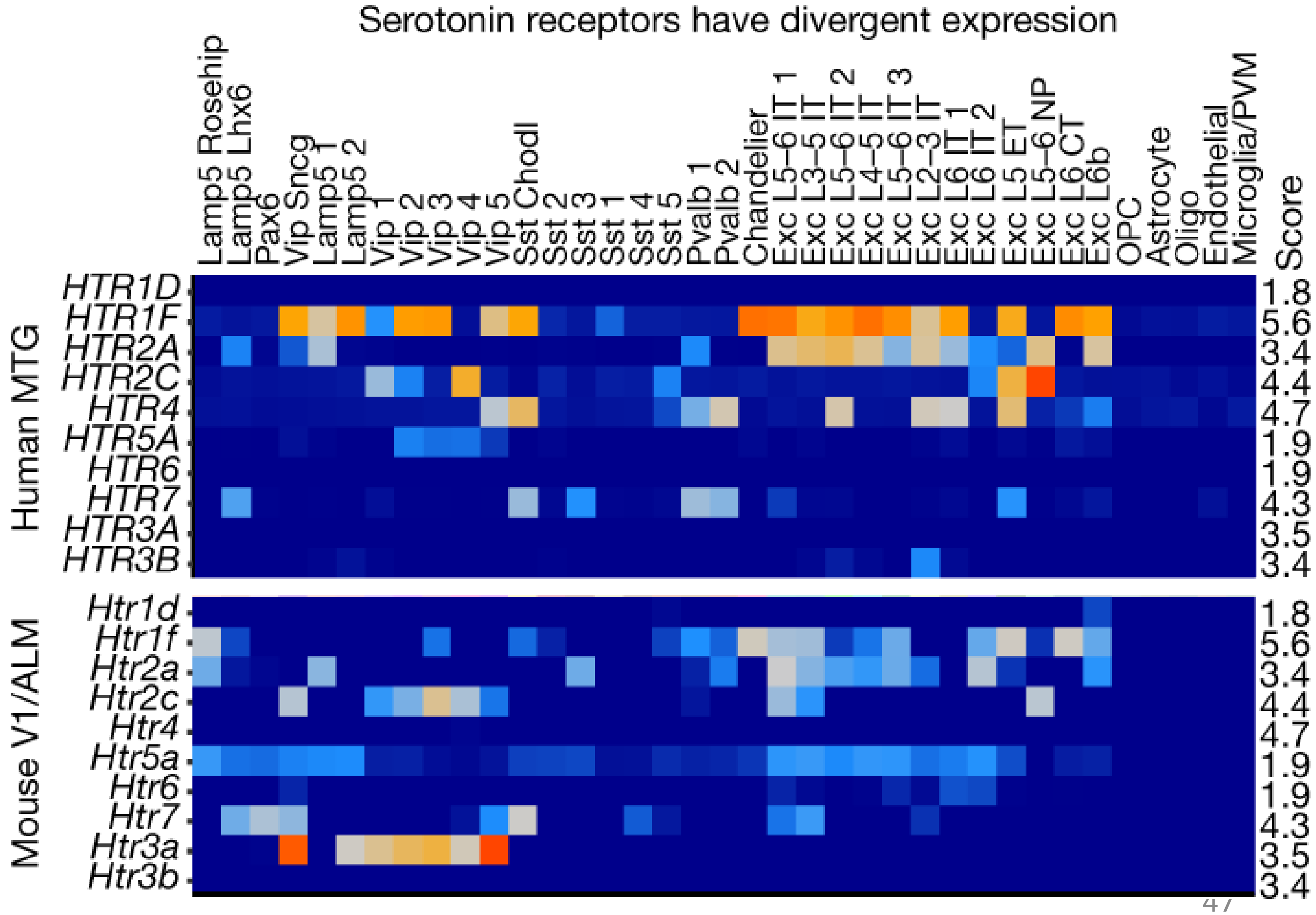




Hodge*, Bakken*, et al., Nature 2019

Divergent cell-type expression between species





Hodge*, Bakken*, et al., Nature 2019

Summary 2

- Single-cell RNA sequencing is very instrumental in building detailed maps of cell types across brain regions, development stages

- These unbiased maps provide a better understanding of the building blocks of the brain and the molecular signatures underlying functional diversity of cells

- Comparing cells across species helps us in understanding evolution and has a direct translational value

Resources

- MGC/BioSB Course - Single Cell Analysis:

https://github.com/mahfouzlab/FOS_Translational_Medical_Genomics_2024

MGC/BioSB Course – Spatial Omics Analysis:

https://github.com/mahfouzlab/MGC-BioSB-Spatial-Omics-Analysis-2025

- Join Single-Cell Network Netherlands (scNL): https://www.singlecell.nl/